

Standards for Emergency Department Design and Specification for Ireland

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Purpose

The Irish Association for Emergency Medicine (IAEM) Emergency Department (ED) Design Guidelines aim to assist clinicians, government officials, health planners, architects and designers in understanding the key infrastructure components of a contemporary Emergency Department in Ireland. The purpose of this document is to:

- ♦ Clearly outline the clinical infrastructure requirements essential for an ED.
- ♦ Emphasize the need to integrate clinical requirements, functional needs and practical size considerations in ED design.
- ♦ Describe the crucial relationships within the ED and its connections to other hospital departments.
- ♦ Highlight elements of ED design that are effective and those that are not.
- ♦ Stress the significance of good design for all newly developed or redeveloped EDs in Ireland, irrespective of location, size or clinical user groups (i.e. doctors, nurses, and other health professionals).
- ♦ Assist in creating EDs that enhance the experiences of patients, families/carers and staff.
- ♦ Foster dialogue and collaboration between ED staff, architects, designers, engineers and health planners, ensuring the practical application of their knowledge and expertise in ED operations.

While the guidelines emphasize the integration of clinical requirements, functional needs, and practical size considerations, they are not intended to serve as an exhaustive guide to models of care delivery and patient pathways. The [appendix](#) includes summaries of functional content along with a sample Schedule of Accommodation detailing all the necessary components found in a standard ED.

Given the dynamic nature of EDs, models of care may change rapidly due to various factors, such as government policies, patient demographics, staffing and other initiatives.

These guidelines should be seen as complementary rather than in isolation to existing guidelines such as the *Health Building Note 15-01: Accident and Emergency departments*.¹ Health Building Notes (HBNs) offer optimal guidance for designing and planning new healthcare facilities and adapting/extending existing structures. These notes furnish information to assist in the briefing and design phases of individual projects within building programmes worldwide.

IAEM regularly produces and reviews policy documents, including these Emergency Department Design Guidelines, to benefit our patients, members and other relevant stakeholders². The insights and knowledge shared in these guidelines are valuable for Consultants in Emergency Medicine who may not have expertise in architecture, design, or planning but should play a significant role in the planning and design process of the Emergency Department in which they practise.

Background

The Emergency Department (ED) plays a vital role in providing acute healthcare access to the public in Ireland, while also offering support to primary healthcare and community services³. It serves as a crucial link to the broader range of inpatient and outpatient services available within its parent hospital and the larger healthcare system/Regional Health Area. The majority of acute admissions to inpatient wards originate from ED attendance, reflecting their essential role in the Irish healthcare landscape.

Characteristics of EDs in Ireland

Several distinctive characteristics of Irish EDs present unique design challenges, including:

- ◆ Existing level of crowding due to lack of inpatient bed capacity.
- ◆ Evolving models of care within EDs, often accompanied by the integration of associated inpatient departments.
- ◆ Varied staffing levels required to support different models of care.
- ◆ Diverse case mix of acutely unwell patients, frequently presenting with time-sensitive and life-threatening conditions.
- ◆ Role differentiation of EDs, such as whether they serve as Major Trauma Centres.
- ◆ Presentation of patients with undifferentiated conditions, necessitating rapid assessment and treatment.
- ◆ Patients and those accompanying them often experience heightened stress and anxiety.
- ◆ Patients presenting with acute psychosocial crises.
- ◆ High patient turnover rates, requiring efficient management and throughput.
- ◆ "Front-loading" of patient care by healthcare professionals to expedite patient flow through the department and hospital.
- ◆ An expanding range of specialised services provided within EDs including frailty services for example.
- ◆ An evident driver of increased attendances is Ireland's growing population, which has surpassed 5 million people, indicating a total growth of 8.1% over the six years since Census 2016, with an annual average increase of 1.3%. Particularly noteworthy is the 25% increase in the number of people aged 85 years and over.

These characteristics underscore the multifaceted nature of ED operations in Ireland and highlight the need for innovative and adaptable design solutions to meet the evolving demands of acute healthcare delivery in Ireland, increasingly reflected in mature healthcare systems internationally.

Good Design Practice

Delivering a new ED requires a holistic approach where design, processes, communication, and adaptability are integral components. The 'form follows function' concept is threaded through this approach, emphasizing the importance of aligning individual design elements with the efficient delivery of Urgent and Emergency Care (UEC) services. The success of a new build or refurbishment is contingent on the effectiveness and efficiency of the processes within the ED⁴.

Design Principles

An aesthetically pleasing design alone cannot enhance emergency care delivery; it must be coupled with streamlined and contemporary processes. The key components of department delivery are:

Processes: The backbone of design

The effectiveness and efficiency of all pathways and processes in the ED must be supported by the new design to enhance emergency care delivery. Outdated or inefficient processes can hinder the success of a new build or refurbishment, emphasizing the importance of integrating form and function seamlessly.

Bridging the gaps

Communication routes are integral to the success of any healthcare infrastructure. This document emphasizes the importance of paying equal attention to both design evidence and communication routes.

Navigating change

Given the resource and management pressures and other challenges faced in EDs, adaptability is key; the ability to change and evolve treatment areas in response to dynamic patient-centred requirements is essential.

Patient-focused principles

The aim of design guidance is to understand best practices, tools and techniques for the wider design team and to provide insights into activity analysis, decision-making and project programming.

User-centric design considerations

Design considerations extend to careful assessments of the needs of patients, staff, and carers/family members during the planning stage. The involvement of these groups ensures individual user needs are addressed, in terms of safety, privacy, security and wayfinding.

Flexible design

Designing an ED is a complex task, requiring careful consideration of several factors, including service requirements, clinical delivery and the inevitability of change. Below we explore the principles for sharing space in ED design, emphasizing the need for large, flexible spaces with multiple components that can adapt to evolving needs.¹

Principles for sharing space

- ◆ Decisions in ED design should be grounded in reliable data and a commitment to continuous improvement. The use of data allows for evidence-based decision-making, ensuring that the space is optimised for efficient patient care.
- ◆ Involving multidisciplinary groups in developing the long-term service vision is crucial. This collaboration enables a comprehensive consideration of spatial limitations, infrastructural opportunities and the capacity to accommodate change over time.
- ◆ Successful change management and resource intelligence involve understanding and mapping uncertainties, challenges and opportunities. A scenario-based approach helps the design team plan for long-term resilience, asking critical "what if?" questions to anticipate and address potential future challenges.
- ◆ Decisions in ED design should consider policy change, service change and clinical change. This includes considering government-driven policies, shifts in service configurations and advances in best clinical practices.
- ◆ The year-on-year increasing attendances at EDs necessitates a thoughtful approach to clinical space design. Understanding why people attend is crucial. Future attendance projections should consider local population growth, changes in infrastructure and the impact of specific incidents or service reconfigurations.

Need for large flexible spaces

- ◆ Large flexible spaces are essential for accommodating the inevitable changes in service requirements and clinical delivery. These spaces allow for seamless adjustments to emerging needs, ensuring the ED remains adaptable and responsive to evolving healthcare demands.
- ◆ Large flexible spaces facilitate scenario planning, enabling the ED to prepare for various potential futures. This adaptability ensures that the physical environment can support the ED's functions under different circumstances, including major incidents and other, less immediate, but as yet unforeseen, challenges such as learnings related to infectious disease management through the COVID-19 pandemic (see section 6.1.8 for recommendations)
- ◆ A large flexible ED layout contributes to long-term resilience by accommodating changes in attendance rates, clinical practices and service configurations. It allows the ED to absorb uncertainties, challenges and opportunities while maintaining operational efficiency.

Patient experience

The design principle of the patient experience in the ED revolves around creating an environment that ensures optimal care, comfort and efficiency for individuals seeking emergency medical attention⁵. Several key elements contribute to this design principle:

Accessibility and navigation

- ◆ Ensure clear and intuitive wayfinding to guide patients to different areas in the ED.
- ◆ Designate separate entrances for walking and ambulance patients to streamline the arrival process.
- ◆ Implement signage that is easy to understand, providing directions to key areas like waiting rooms, treatment spaces and restrooms.

Patient-centred spaces

- ◆ Create waiting areas that prioritise patient comfort, with comfortable seating, positive distractions (such as art or entertainment) and access to refreshments.
- ◆ Design treatment spaces with a focus on privacy, while still allowing for supervision by staff when necessary.
- ◆ Integrate areas specifically tailored to the needs of particular patient populations, such as children and older people, those with bariatric issues, sensory issues and those with certain other conditions such as pre-existing cognitive or mobility issues.

Efficient triage and admission

- ◆ Develop a well-organised triage area that allows for early assessment and prioritization of patients based on the severity of their condition.
- ◆ Design admission processes that minimise wait times and facilitate smooth transitions from triage to treatment areas.

Communication and information

- ◆ Implement effective communication systems, e.g. nurse call panels, to ensure that patients can easily request assistance.
- ◆ Support the provision of clear and concise information about the patient's status, treatment plan and expected wait times to reduce anxiety and uncertainty.

Comfort and wellbeing

- ◆ Consider factors like noise control, lighting and air quality to create a healing and calming environment.
- ◆ Ensure the availability of comfortable seating and spaces for patients, older people, children and their families.

Safety and security

- ◆ Implement security measures to safeguard patient privacy and ensure the safety of both patients and staff.
- ◆ Designate secure areas for patient valuables and belongings.

Technology integration

- ◆ Integrate technology to enhance patient care, such as electronic health records (EHRs), telemedicine capabilities and efficient diagnostic tools.
- ◆ Ensure that staff has access to necessary technology for quick and accurate decision-making.

Flexibility and adaptability

- ◆ Design the ED with flexibility to adapt to changing patient volumes, diverse medical needs and emerging technologies.
- ◆ Plan for expansion or modifications based on future healthcare trends and advancements.

KPIs and benchmarks

The implementation of key performance indicators (KPIs) and benchmarks in an ED is a crucial aspect of designing models of care that prioritise the provision of high-quality and safe healthcare. The development of these models involves collaboration among clinicians, non-clinical practitioners and support staff to ensure patient pathways align with recognised best practices and standards.

Resilience

Resilience stands as a cornerstone in the effectiveness of an ED, serving as the frontline interface of a hospital with the wider community and other healthcare providers and ensuring uninterrupted service provision, particularly during major incidents. A critical aspect of this resilience lies in the department's capacity to handle day-to-day fluctuations and sudden surges in patient influx during emergencies, a consideration integral to the design process. Spatial and operational planning should extend its adaptability to withstand extreme pressures, especially in catastrophic events, intertwining major incident and resilience planning with features like decontamination and infection control.

The resilience plan for an ED needs to seamlessly navigate through these key stages of escalating acute need: from routine operations to operational adaption, departmental adaption to organisational adaption for crisis measures.

This strategic planning ensures that the system anticipates shifts beyond routine, initiating resource changes, department-wide system changes and planned reorganisation to sustain operations during unprecedented events.

As part of adaptability and future-planning, the ED's design strategy should embrace future-proofing to accommodate expanding needs with service growth. This involves a comprehensive evaluation of long-term costs, including repair and replacement cycles, and staffing requirements, thereby mitigating financial and operational risks associated with a lack of adaptability.

To gain insights into future needs, the design team is encouraged to conduct future-planning workshops with stakeholder groups, adopting a scenario planning approach. This involves assessing potential changes over five, ten, and 20 years, allowing the ED to proactively respond to evolving requirements. Drawing from lessons learned from past service requirement changes, accurate predictions and the costs of not meeting these changes, the multidisciplinary team should engage in reflective questioning such as "What if?" and "What happens if?" to inform a comprehensive and adaptive ED design.

Digital

The integration of Information and Communications Technology (ICT) in EDs marks a crucial milestone in healthcare.

This section explores how clinical innovation coupled with evolving technologies can revolutionise operational processes, streamline patient care, and ensure efficiency within EDs.

Clinical innovation serves as the driving force behind operational changes in EDs. One critical aspect is the emphasis on rapid initial working diagnosis, assessment and treatment. Proximity to diagnostic equipment and efficient processes reduces unnecessary admissions, minimises handovers, and mitigates the risk of infection exposure.

The landscape of EDs is evolving with the advent of new technologies, particularly in telemedicine and mobile imaging. Bed management systems that enable effective coordination and capacity management, electronic noticeboards connected to ambulance/paramedic networks facilitate early patient assessment, thereby enhancing overall operational efficiency are examples.

The implementation of virtual wards, leveraging telecommunication management, emerges as a significant trend aimed at avoiding unnecessary admissions. Furthermore, ICT plays a pivotal role in streamlining front-end processes, effectively communicating waiting times to patients and enabling self-check-in.

A key consideration in optimizing EDs for the digital age is the compatibility of ICT systems within the department and the broader hospital infrastructure. Unrestricted access to radiology and patient test data improves decision-making times for admission or discharge.

The multidisciplinary design team must proactively anticipate emerging technologies, ensuring that spaces align seamlessly with evolving requirements. Strategic divestment in non-digital infrastructure, coupled with unified mobile communication devices for staff, marks a paradigm shift.

Sufficient computer access in administrative and office areas, along with secure electronic information access, enhances staff efficiency. Anticipating the shift toward electronic medical records, minimal storage for paper records, and innovative technology integration for wayfinding are paramount.

Common pitfalls include inadequate space anticipation for the growing electronic medical resources, treating IT as a separate project, neglecting early IT implementation, and infrastructure gaps that fail to account for future trends. Acknowledging the shift to fully digital electronic records, the current trend emphasizes ensuring network points at every patient treatment space and the move towards a computer per cubicle or the use of handheld tablets (see 6.2.2 for recommendations). An appropriately climate-controlled room for charging handheld Acute Floor Information System (AFIS) devices is therefore a requirement in EDs in Ireland into the future.

Standardisation

In the realm of Emergency Department (ED) design, standardisation emerges as a key principle with far-reaching implications. A consistent approach to room-handling and layout has proven to be a powerful tool in minimising medical errors. This design strategy advocates for uniformity in the placement of doors, their direction of opening, and the positioning of essential elements such as equipment, communication points, and mechanical/electrical switches. Such consistency is not merely about aesthetics but is fundamental for ensuring safe and effective clinical care.

The benefits of standardisation extend beyond reducing errors. Increased standardisation of supporting services, such as medical gases, contributes to the flexible use of rooms catering to a broader range of need. Performance specifications, like solid partitions between rooms/bays and corridor entrances, not only boost patient privacy but also enhance environmental performance and operational resilience.

Health Building Notes (HBNs) and other guidance documents provide much of the determination on the size of core treatment areas. Determining other area's sizes becomes a strategic decision, informed by analysis of activity flows, clinical requirements, and the operational use of technology and mobile equipment. Room capabilities and sizes are influenced by the desired level of patient experience and organisational flexibility as well as clinical demands.

Integration of design with operational strategies to create ED environments that prioritise patient safety, privacy, and effective clinical care is the goal.

Governance

When designing an ED, effective leadership and governance, including frontline clinical leadership together with estates, health planners, architects, designers, engineers and other key stakeholders, is crucial. This will likely include in-house Specialty Services and healthcare groupings like nursing, administration, portering, household, catering, IT, doctors in training, HSCPs, Ambulance Service, etc.

Performance challenges like those set out in HIQA reports and Department of Health Reports ^{7,8} should be taken into account.

Ensuring ED crowding is recognised as a whole health system shared problem with zero tolerance for normalizing comparatively poor performance is vital.

Care Models and Pathways

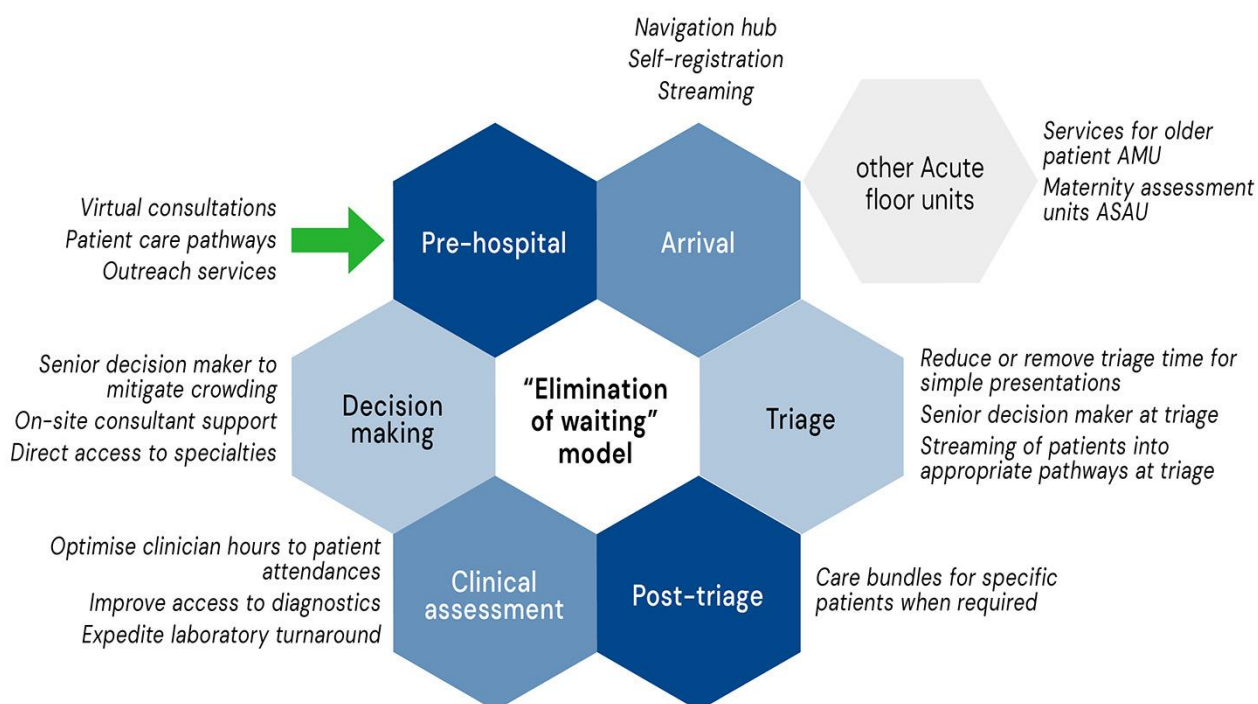
The Big Front Door

The "Big Front Door" concept derives from this integrated approach to accessing hospital services and broader local health service provision. It's success hinges on adopting a comprehensive system perspective to establish an effective and efficient delivery mechanism, spanning preventive medicine to social support frameworks. An integrated "front door," coupled with effective bed space management, has the potential to mitigate admissions, shorten length of stay and enhance recovery times. A key principle is early up front senior clinical decision-making facilitating streaming to key areas within the ED and the acute floor or to associated facilities like injury units, urgent care centres and, associated GP services that may be co-located.

The Acute Floor model

The Acute Floor Model in Ireland offers an opportunity to facilitate streaming to acute medicine, acute surgery, frailty, cardiology etc. This design focuses on strategic patient flow, rapid streaming and avoiding unnecessary hospital admissions.

Figure 2 - Acute floor model



Pre-hospital

The design would involve the following elements:

Integrated triage and pre-hospital assessment

- ♦ Ambulance services collaborate closely with the ED to conduct pre-hospital assessments.
- ♦ Triage protocols are aligned to the Acute Floor Model, allowing for streaming and patient categorization based on urgency and required expertise.

Advanced communication and information sharing

- ♦ Enhanced communication systems between pre-hospital providers and the ED facilitate real-time information exchange.
- ♦ Relevant patient data, such as medical history and current medications, is transmitted to the ED, aiding in better preparedness for incoming patients.

Mobile technology for decision support

- ♦ Pre-hospital personnel will increasingly use mobile technology for decision support, allowing access to protocols, guidelines and real-time consultation with ED specialists.

Acute Medical Assessment Units

Acute Medical Assessment Units or AMAUs are predominantly found in Model 3 and 4 Hospitals, whereas Medical Assessment Units are usually situated in Model 2 settings. AMAUs serve as alternative pathways into acute hospital services, distinct from EDs. AMAUs can play a role in the acute care system offering alternative pathways and streaming opportunities.

When building an ED with an associated AMAU, it is essential to consider key performance indicators (KPIs) related to hospital efficiency, particularly those associated with admissions procedures.

When designing an ED, careful consideration of the physical layout, staffing, and operational processes is necessary to optimise the utilization and effectiveness of AMAUs.

Ref: https://www.iaem.ie/wp-content/uploads/2013/07/report_of_the_national_acute_medicine_programme_2010.pdf

Same Day Emergency Care (SDEC)

Same Day Emergency Care (SDEC) has been implemented across the NHS and other countries to improve patient care by providing swift investigation and treatment, ensuring non-elective patients, who traditionally would have been admitted, receive comprehensive care within a single day. The Health Building Note (HBN)15-02: *Facilities for same day emergency care/ambulatory emergency care* offers guidance on designing an SDEC department, emphasizing a well-planned layout to manage patients effectively and reduce delays in their care journey⁹. The document outlines the patient pathway, essential adjacencies with departments like the ED and functional relationships with diagnostic imaging.

Core recommendations include proximity to an ED, prompt access to diagnostics, and a combination of consulting rooms, patient trolleys, and chairs for assessments. Excluding in-patient beds, SDEC treats a range of conditions, streamlining referrals from various sources, including ED triage, direct GP or ambulance referrals.

The design incorporates flexible treatment options, such as acute medical, surgical, and acute frailty services. The NHS has outlined goals for SDEC availability and acute frailty services, emphasizing the importance of a paperless system for efficient data management. Patient streams, including diagnostic exclusion, procedural, prolonged treatment, and ad-hoc groups, guide the SDEC unit's design, ensuring flexibility and responsiveness to local policies and patient needs.

Built Environment

Accessibility

It is important to establish distinct entrances for pedestrians and ambulance arrivals. These entrances should be exclusively designated for ED access and should never serve as general entry points to the hospital or pathways for staff and non-ED patients traversing within the facility. Such usage can adversely impact ED patient care, compromise privacy, and potentially jeopardise the safety of both patients and staff.

Furthermore, the clinical areas within the ED must remain off-limits to medical, nursing, or other staff not engaged in specific patient care activities within the ED. This approach ensures a focused and secure environment tailored to the unique needs of ED patients.

Designated free parking areas adjacent to the Emergency Department should be assigned for both disabled individuals and caregivers with infants and young children.

Lighting

Ambient lighting in Emergency Departments (EDs) aims for a calm atmosphere, mimicking natural light patterns.

Task lighting in clinical areas is essential, with a target of 30,000 lux. Patient treatment spaces benefit from dimmable lighting, calming colour schemes, and controlled ambient temperatures with air conditioning.

Increasing daylight to EDs and designing to accommodate available natural light can also impact on energy consumption. It is also likely to improve staff well-being. See also CIBSE's (2008) *Lighting Guide LG2 – Hospitals and health care buildings*.

Wayfinding

For any hospital with an ED, effective signage on the hospital campus is crucial for accessibility. The local authority and/or Transport Infrastructure Ireland should oversee clear signposting from major roads entering the area.

All entrances to the hospital campus should feature explicit signs guiding to the ED, differentiating between ambulance/vehicular and pedestrian access. Internal hospital corridors and thoroughfares must also have clear signage directing to the ED, using the convention of "Emergency Department" in white lettering on a red background for consistency¹⁰.

Within the ED, the wayfinding system should align with the hospital-wide system, employing the same styles and conventions. It is recommended to place signs at major intersections and destinations. In the absence of intersections, signs should confirm the route approximately every 30 meters. Electronic wayfinding and information kiosks are increasingly common.

Given infection control concerns, some EDs may opt for wall-based wayfinding directions instead of floor lines. Prominent room signs for toilets, including braille for visually impaired patients, and hearing loop or infra-red systems for those with hearing aids are essential. Pitfalls to avoid include major corridors ending abruptly without staff bases or 'touch points' for proper guidance through the ED for patients and visitors.

Privacy

Privacy in the ED is prioritised through the design of individual cubicles. Ideally, all ED cubicles should be single occupancy. Each cubicle, at a minimum, must have three walls and a curtain to ensure patient privacy. Some specific cubicles, particularly those designated for paediatric patients, migraine cases, and gynaecological procedures, should have enhanced privacy features such as walls or robust partitions along with a door to not only safeguard privacy but also to attenuate noise.

Infection control standards and hand hygiene

Hand wash basins equipped with hands-free taps should be conveniently located near treatment areas, patient cubicles, and specific areas such as resuscitation rooms, point of care testing areas, dirty utility rooms, and the staff room [9].

For situations involving non-soiled hands, alcohol-based hand rubs (ABHRs) serve as an effective alternative to detergents, reducing rates of dermatitis. These ABHRs should be easily accessible, whether mounted on walls, trolleys, or similar fixtures, at all ED entrances and within all clinical areas. However, it is essential to exercise caution in the placement of ABHRs, particularly in areas with paediatric patients or individuals with impaired judgment or tendencies towards self-harm, as they may be toxic if ingested in significant quantities [10].

Total size of Emergency Department

The size of the ED is contingent on service volume and diversity. Various factors influencing overall floor space include attendance patterns, patient acuity, length of stay, admission rates, turnaround times for diagnostic investigations, age distribution, academic activities, paediatric care requirements, and mental health patient numbers. Ample space provision should accommodate evolving work patterns, multidisciplinary team collaboration, changing service needs, an aging patient population, new medical interventions, and monitoring demands.

Anticipated occupant loads, based on International Occupant Load formulas, should be considered, ensuring compliance with fire regulations and health and safety requirements. The design and size of new EDs should adhere to outlined requirements without compromise, despite existing substandard infrastructure, as it directly impacts staffing and operational procedures.

Calculating space requirements for the ED

To understand how to calculate for space allocation in an ED, it is essential to grasp the demographics and acuity levels of patients who attend; to recognise the projected population growth in the area served and incorporate healthcare trends (for example the aging population, changes in Primary Care unscheduled care delivery etc.) into planning.

Demand and capacity modelling is an approach used to forecast future capacity requirements for healthcare services, for example resus, majors and ambulatory spaces. A detailed demand and capacity analysis is done with health planners, architects and designers, the hospital's estates personnel / hospital engineers and the senior leadership in ED and the hospital. This should include future modelling based on demographic and non-demographic changes, and will be completed utilising existing data sources, along with experience and understanding of best practice and emerging trends.

To calculate space allocations, we make various assumptions about operational aspects:

- ♦ Operating Days: The number of days a facility operates within a given time period.
- ♦ Operating Weeks: The number of weeks a facility operates within a given time period.
- ♦ Occupancy Rate: The percentage capacity at which each section of the ED should operate.
- ♦ Treatment Minutes: The average time required to treat a patient in a specific section.
- ♦ Peak Period Hours: The number of hours with the highest percentage of attendees throughout the day.
- ♦ Peak Period Proportion of Activity: The percentage distribution of patient activity in these peak period hours.

Below is a hypothetical table illustrating how these factors contribute to space requirements for 1000 attendees in each treatment zone:

	Modality	Ambulatory	Minors	Majors	Resus	Paediatrics
Assumptions	Anticipated Annual Attendees	1000	1000	1000	1000	1000
	Operating Days	7	7	7	7	7
	Operating Weeks	52	52	52	52	52
	Occupancy Rate	65.00%	65.00%	65.00%	45.00%	60.00%
	Treatment Minutes	45	60	160	200	90
	Peak Period Hours	10	10	11	10	10
	Peak Period Proportion of Activity	75.00%	75.00%	75.00%	75.00%	80.00%
Calculations	Peak Period Hours Per Year	3640	3640	4004	3640	3640
	Treatment Hours	562.5	750	2000	2500	1200
	Total Treatment Spaces	0.2	0.3	0.8	1.5	0.5

These assumptions are in line with best practices and are subject to adjustment based on the operational targets established at an individual hospital. Variation will be most seen in relation to minors and ambulatory care, which will orientate itself differently depending on the acuity of attendees and location. Based on these assumptions, a high level calculation to demonstrate total treatment spaces, looks like this for the example above:

A = Occupancy Rate

B = Operating Days

C = Operating Weeks

D = Peak Period Hours

E = Anticipated Annual Attendees

F = Treatment Minutes

G = Percentage Peak Period Proportion of Activity

$$\text{Total Treatment Spaces} = \frac{A}{(B \times C \times D) \times \left(\frac{E \times F}{60 \times G}\right)}$$

In a comprehensive analysis, factors beyond demographics, such as left-shift initiatives, patient acuity/specialty, and services like Rapid Assessment and Treatment (RAT) and AMAU, would be included to provide greater accuracy in estimating treatment space requirements. This analysis would also consider future capacity needs. Furthermore, there would be extensive stakeholder engagement to understand potential reorganization.

Limitations

- ♦ **Patient Re-allocation:** The current approach lacks accounting for potential patient re-allocation, which could lead to inaccuracies in space allocation planning e.g. if a patient moves from minors to majors, or other intra ED moves.
- ♦ **Lack of Sensitivity Analysis:** Without conducting sensitivity analysis, the current model may not capture variations in demand, compromising the accuracy of space allocation projections.
- ♦ **Future Demands Oversight:** Future demand modelling needs to incorporate emerging trends such as left-shift initiatives, digital demands, and clinical team decisions to ensure accurate space allocation planning.
- ♦ **Variable Definitions of Care Areas:** Care areas may have variable definitions across different hospitals. For instance, while ambulatory care in some hospitals includes RAT or Rapid Assessment Streaming Triage Treatment Area (RASTTA) it may not always exist. Additionally, the calculation of chair spaces and sub-wait areas does not follow a standardized method but often relies on expert opinion for space allocation depending on acuity and definition of activity occurring in different zones. This approach enhances space utilization efficiency, thereby leading to a more positive patient experience.
- ♦ **Location and Access:** Considerations such as geographical perspective, location within the hospital, and availability of other services should be factored in.

Net zero carbon emissions

Healthcare services play a substantial role in contributing to carbon emissions, with estimates suggesting it accounted for approximately 5-15% of greenhouse gases in 2019. In Ireland, healthcare organisations share a responsibility to address this environmental impact by aligning with Net Zero Carbon targets and implementing broader sustainability measures.

The HSE is committed to achieving net-zero emissions no later than 2050, delivering healthcare which is environmentally and socially sustainable as set out in its strategy *HSE Climate Action Strategy 2023 – 2050*¹¹.

Service panel

In the context of ED design, service panels play a crucial role in providing power supply for equipment, medical gases, suction, and bedside communications.

General Power Outlets (GPO)

Functional requirements for service panels, as advised by the Australasian College for Emergency Medicine (ACEM), in different areas of the ED are as follows:

Resuscitation room (for each patient space)

- ◆ 3 oxygen outlets
- ◆ Medical air outlet*
- ◆ Nitrous oxide outlet with scavenging unit*
- ◆ 3 suction outlets
- ◆ 16 General Power Outlets (GPOs) in at least two separate panels (at least 50% on essential power)
- ◆ 6 data outlets:
 - » Staff call panel
 - » Allows for physiologic monitor networking plus remote monitor
- ◆ Includes phone and computer

Acute treatment bed - adult and paediatric

- ◆ 2 oxygen outlets
- ◆ Medical air outlet*
- ◆ 2 suction outlets
- ◆ 8 GPOs in two separate panels (at least 50% on essential power)
- ◆ Nitrous oxide outlet with scavenging unit*
- ◆ 4 data outlets:
 - » Allows for physiologic monitor networking
 - » Staff call panel
 - » Phone (optional) and computer

Procedure room/suture room/plaster room

- ◆ 2 oxygen outlets
- ◆ 2 suction outlets
- ◆ 8 GPO in two separate panels (at least 50% on essential power)
- ◆ Nitrous oxide outlet with scavenging unit*
- ◆ 6 data outlets:
 - » Staff call panel
 - » Allows for physiologic monitor networking
 - » Includes phone and computer with PACS (Picture Archiving and Communication System) access

Consultation room

- ◆ 1 oxygen outlet
- ◆ 1 suction outlet
- ◆ 4 GPOs (at least 50% on essential power)
- ◆ 4 data outlets:
 - » Allows for physiologic monitor networking
 - » Staff call panel
 - » Includes phone/computer

External service panels

- ◆ 3 oxygen outlets
- ◆ 2 suction outlets
- ◆ Staff call panel
- ◆ 12 GPOs in at least two separate panels (all on essential power)
- ◆ Nitrous oxide outlet with scavenging unit*

* item is optional depending on the model of care/equipment used.

Functional Content

Reception

The reception area in an ED plays a crucial role in patient care by electronically recording personal details, essential for generating necessary paperwork and patient labels during the current episode of care. It facilitates the retrieval of hard copy medical records, streamlining the administrative process and ensuring seamless continuity in patient management

Size

- ♦ The minimum acceptable floor area for a Reception is 12m²
- ♦ The size is determined by the patient census, dictating the number of required reception workstations.

Design considerations

Reception area and patient records	Each ED has a dedicated reception area staffed 24/7 by trained personnel. Immediate access to ED records for the previous two calendar years is required, especially if not electronically stored.
Patient confidentiality and security	Protect patient confidentiality during registration, with emphasis on security for reception staff. Utilise appropriate communication systems to balance confidentiality and security needs.
Waiting area design	Ensure waiting patient seats are at an appropriate distance from the reception desk. Consider separating the reception area and waiting areas for efficiency. Provide a wheel-chair accessible reception booth and address the needs of disabled patients, including those with hearing and vision impairment. Accommodate patients who may need to sit during registration due to injury, illness or disability. Consider registration facilities for patients arriving by ambulance in the overall design. Consider self-registration terminals
Administrative function and design layout	Situate the department's main administrative function in a "non-clinical" area separate from reception. Ensure that only functions directly related to clinical care are located in the most operationally active zones in the department.
Other	Patient Registration should take place before, or at the same time as triage. Delays to triage must be minimised

Functional content

Spatial relationships

- Relationship with retrieval of hospital medical records if they are not electronic
- Relationship to a Triage Nurse, either as a triage and a registration pairing, or a separate area for registration only (following triage)
- Relationship to the entrance of the hospital after hours
- Relationship to the ambulant entrance of the ED (often Registration Clerk is responsible for regulating entry and exit to the ED)
- The design must allow for clinical staff to have a clear line of sight into the reception area.

Functional requirements	
• Dedicated registration area co-located with triage • Access to an electronic patient information entry portal • Mechanisms to ensure privacy for disclosed patient details and/or displayed patient information	• Security of staff from any aggressive patients • Duress alarm • The use of mobile registration at the bedside using mobile devices or bedside computer terminals optimal.
Specific equipment requirements	
• A comfortable and ergonomic workspace • Printer • Forms and stationary storage	• Computer workstations (may be portable) with phone and other forms of communication.

Common design flaws

- ◆ Not having enough space for staff
- ◆ Temperature control, especially the effect of computers, printers, and glass.
- ◆ Wheelchair and disability accessibility not accounted for and how this affects administrative office and engagement with clinical staff etc.

Waiting room

The waiting room in an ED serves as a critical space for patient observation, ensuring both clinical oversight and security. It aims to minimise patient agitation through thoughtful design, providing comfort, amenities, and information. The waiting room functions as a hub for communication, offering distractions, and facilitating efficient patient flow with, where possible, electronic systems for real-time updates on waiting times.

Size

- ◆ Number of waiting spaces will be calculated in proportion to treatment areas and upper range of attendees.
- ◆ The flow of the department (patients, staff, logistics) needs planning at an early stage to ensure the right mix of waiting spaces, and the right locations. Good practice dictates for average departments about 3 chairs per actual cubicle/activity space to recognise the requirements.
- ◆ The audio-visual separation of Paediatrics is crucial. Further space must also be added for disability groups.
- ◆ The waiting area allowance is 1.5m² per seat.
- ◆ Ensure a minimum of three seats per patient treatment cubicle.
- ◆ Infectious disease considerations should also be taken into account.

Design considerations

Waiting room design	The waiting room should prioritise patient comfort, safety and wellbeing. Consider the waiting room as a clinical area overseen by ED staff. Ensure good communication with waiting patients to prevent feelings of isolation. Observation of waiting patients for clinical and security reasons. Staff should be aware of waiting times and address patient requirements for refreshment during prolonged waits. Minimise patient agitation through thoughtful design, including appropriate artwork, lighting and seating arrangements.
Information and amenities	Maintain controlled access between the waiting room and the rest of the ED. Allow access to the waiting room from the treatment and ambulatory areas. Facilities for patients and accompanying individuals (see 6.1.3 Refreshment supply zone) Provide toilet access and baby changing facilities. Include a direct line to a taxi service.
Paediatric waiting area	Design a larger waiting room to accommodate parents likely to have buggies etc. Establish a separate supervised paediatric waiting area with a play area. Use laminated one-way view glazing for privacy, allowing people to see in without children seeing out. Locate toilet facilities, baby change and infant feeding close to the children's waiting area.
Information and waiting experience	An electronic display with current waiting times. Sub-waiting areas within the ED. Ensure access to health-promoting materials. Facilities for charging mobile phones and electronic devices. Electronic systems that allow patients to be in other areas of the hospital while waiting and then be recalled when they can be seen by a clinician.

Functional content

Spatial relationships	
• Triage • Ambulance and walk-in entrance	• Reception • Clinical areas of ED.
Functional requirements	
• Ambient temperature control • Access to food, drink, and toilets • Special accommodations for individuals with disabilities, children, and mothers, including feeding and nappy change facilities and secure play areas • Patient information, signage, indication of wait times, access to written material and public health information • Distractions during the wait, such as public television, magazines and video entertainment for children	• Emergency call system (patient-activated in toilets) • External vehicle drop-off area and provision for short-term parking • Communication facilities, including public phones and free taxi phones. Facilities for individuals with disabilities, including those in wheelchairs or with mobility, sight or hearing impairment. • Adequate signage, wayfinding and seating to accommodate mobility aids and wheelchairs • Colour-coded seating for zoning.

Specific equipment requirements

- | | |
|---|--|
| <ul style="list-style-type: none">• Portable patient monitoring and assessment equipment• A security system including monitored CCTV, emergency call system including duress alarm for staff | <ul style="list-style-type: none">• Display system for patient and public health information• Seating and fixtures that cannot be used as weapons in the event of an altercation. |
|---|--|

Common design flaws

- ◆ Inadequate wayfinding
- ◆ Waiting room that is too small
- ◆ Inadequate distractions
- ◆ Oppositional seating
- ◆ A poor public announcement system that cannot be heard
- ◆ Door swings that become hazardous or automated doors that are unduly sensitive to passing pedestrian traffic
- ◆ Lack of segregation of family and children's space.

Ambulance access

Used for delivering both ambulant and non-ambulant (trolley-bound) patients to the ED.

Size

The minimum acceptable floor area for both the ambulance entrance and the ambulance handover area is 12m² each.

Design considerations

Size allowance	There must be an allowance for an unloading space at the rear of the vehicle's side door opening and pedestrian access along both sides of a parked vehicle Adequate circulation space is required at the rear of parked ambulances in order to manoeuvre stretchers. Sufficient space is required in front of parked ambulances to enable manoeuvring of vehicles consistent with the turning circle of such vehicles. Lifting should be minimised by the provision of a plinth or a similar system if drop-down stretchers are not used by the local ambulance service.
Ambulance access and patient flow	Provision for multiple ambulances to park near the ED entrance for safe patient disembarkation. Designated reception/triage area for ambulance patients, controlled by security personnel. Two entrances: "main Emergency Department entrance" for those arriving independently and "ambulance entrance" for ambulance patients. Ideally, no overlap between patients accessing the main entrance and the ambulance entrance. Canopied drop-off zones for weather protection, well-lit. Ambulance parking bays close to the entrance for swift clearance. Consider proximity to triage. "Red Bay" could be located closest to the ambulance entrance to accommodate cases of cardiac arrest, major trauma and immediate offloading of critically unwell patients.
Collaboration with emergency services	Design in collaboration with Ambulance Service, considering turning circles, ambulance cleaning, major incident planning and equipment storage. Ambulance bay serving multiple functions: delivery, decontamination, triage, treatment for Mass Casualty Incidents (MCIs), parking and emergency access for other services (e.g., Police, Fire Brigade). External decontamination facilities may need to be factored into the design. Consider IT developments for potential pre-hospital service space within the ED footprint.
Infrastructure and security	Level and unobstructed access to both entrances. Wash-hand basins at the entrance. Wheelchair storage near the entrance, avoiding corridor or entrance space. Security measures without compromising department security.
Specialised facilities	For bariatric patients, weigh scales and patient lifting devices should be considered. Possible route for relatives with ambulance patients, leading to patient registration or waiting area.

Functional content

Spatial relationships	
• Direct access to triage or a dedicated ambulance triage area • Ideally, direct access to resuscitation rooms • Access to ambulance dedicated store room with cleaning facilities	• Ambulance Entrance proximity to the ED Resuscitation Room for minimal transfer time. • Direct, private, and well-illuminated route to Resuscitation facilities, minimising corners
Functional requirements	
• Dedicated and separate ambulance vehicle access • Signage including ground marking for vehicle bays • Adequate lighting	• Ability to off-load and load patients safely • Ability to transfer patients rapidly, with ready access to resuscitation rooms in time critical situations.
Specific equipment requirements	
• Non-slip surfaces • Adequate drainage system • Access to wheelchair • Airlock on ambulance entrance with controlled entry	• Anti-soiling systems on entrance for foot and wheel contamination • Trolley wash • Hand wash facilities.

Common design flaws

- ◆ No ambulance parking provided to allow non-offloading vehicles to clear the drop-off zone
- ◆ An ambulance entrance that is not exclusive or controlled
- ◆ Inadequate drop-off and parking facilities for ambulances, Garda cars and hospital transport
- ◆ No collaboration with local ambulance personnel during design.

Ambulatory care / consultation rooms

This area would be used to facilitate safe, immediate management in case of unexpected deterioration or re-designation during a Major Incident. For flexibility, these rooms can also be used for counselling or video conferencing.

Size

- ♦ The minimum acceptable floor area per consultation room is 16m²
- ♦ Size of 5m² chair per attendee should be allocated in the sub-wait area.
- ♦ Total size of area dependent on patient volume and treatment approach.

Design considerations

Supporting facilities	Incorporate a sub waiting area for patients. Strategically place a nursing station for efficient workflow. Ensure ease of access to the Diagnostic Imaging Department and Procedure Room from this area.
Spatial layout	Consider local operational procedures and staffing resources to determine whether to integrate this area with the main treatment area or physically separate it by doors or a corridor. Evaluate congruency with the main treatment area or physical separation based on overall department efficiency and safety considerations. A mix of rooms, treatment chairs (for those well enough and likely going home), as well as sub-wait seating areas to reduce the number of patients remaining in the Waiting Room.

Functional content

Spatial relationships	
Should be close to RAT and triage area. Consider proximity to the diagnostics suite, for example, to CT for acute stroke cases.	Be accessible to Procedure Rooms Proximity to isolation facilitates for high risk cases
Functional requirements	
Some rooms located close to entrances for immediate assessment Lockable rooms with external emergency override and visual occupancy indicator At least one room equipped for ENT, ophthalmic, and dental problems. Private room for gynaecology/intimate examination	Isolation suite at the front door for rapid segregation of potentially infectious patients Mixture of Chairs and Recliners. Ensure all attendees have at least a seat.
Specific equipment requirements	
2 oxygen outlets Wall mounted suction A service panel with ten electrical sockets and four Category 5 sockets A patient trolley	Computer terminal and diagnostic image viewing facilities A surgical light A wall-mounted ophthalmoscope and otoscope.

Patient chair	Access to dressings and wound care equipment
Swivel chair/stool	A panic button
Hand-washing facilities	Sharps bin
Patient documentation equipment (writing boards/desk)	Clinical waste disposal bin

Common design flaws

- ◆ Inadequate footprint resulting in limited circulation space and space for IV trolleys
- ◆ Inadequate lighting and lighting that cannot be dimmed
- ◆ Limited privacy – three sides of cubicle curtained; conversations able to be overheard by adjacent patients
- ◆ A lack of correct disposal bins for infectious and non-infectious waste
- ◆ Should have facilities for telemedicine.

Triage

A triage area is designed for the initial clinical assessment of patients and allocation of a clinical priority. It has a single point of entry for acute patients, and it provides controlled access to treatment areas and to the wider hospital.

Size

- ◆ Minimum acceptable floor area per Triage/Assessment Cubicle is 12m² with a minimum of 2
- ◆ For flexibility, consider 16m² as the additional space facilitates greater case mix.

Design considerations

Models of care / patient pathways in triage:	Transition from traditional triage to dynamic assessment and simultaneous treatment. Different pathways for ambulant, non-ambulant, paediatric and mental health patients. Tailored care models for diverse patient needs including mental health, sensory issues and addiction services. The adoption of models of care incorporating medical triaging and Rapid Assessment and Treatment (RAT) systems will necessitate an appropriate increase in size in order to accommodate these additional functions Consider proximity to the ambulance bay. All triage bays should be sized to accommodate ambulance trollies if necessary. It's important to establish a one-way flow through triage cubicles - one door in and one door out. The exit door should ideally lead the patient forward into the "Post-triage space" where enhanced triage, point-of-care diagnostics and/or treatment can occur.
Timing and locations for triage	Triage may occur before, instantaneously upon patient arrival (pre-hospital notification), within minutes of arrival, at the bedside or in a designated area. May occur before or after patient registration based on local practice.
Safety measures in traditional triage	Ensure two exits from each triage cubicle for the safety of assessment nurses. Each triage cubicle equipped with a dedicated alarm.

	Design considerations to ensure the safety of assessment nurses during patient assessment.
Triage / assessment area design	Designated areas for ambulance and ambulant patients. It has a single point of entry for acute patients and provides controlled access to treatment areas and to the wider hospital. Adequate facilities to support dynamic assessment and simultaneous treatment. Preserve patient privacy and confidentiality.
Workflow and access in triage	Access to ECG and diagnostics within the triage area. Immediate drug therapy provision (e.g. analgesia). Administrative functions like printing patient name-bands or initiating patient tracking at triage along with registration are easily available.
Prioritization	Prioritization based on clinical acuity for effective patient care through Manchester Triage Score (MTS).
Patient movement considerations	Movement considerations between Triage/Assessment and reception, waiting area, resuscitation, treatment & ambulatory care, patient toilets, and diagnostic imaging. Ambulatory ambulance borne patients may be streamed to here.
Trolley consideration	Department practice determines whether a trolley is required in each triage/assessment cubicle. Trolley may be needed for RAT or immediate ECG testing for ambulatory patients.
Paediatric-friendly design	Initial assessment of babies, children, or adolescents with non-life-threatening injuries or illnesses in triage/assessment cubicles. Some spaces suitably decorated, furnished, and equipped for children. Specific rooms for babies and children, but all initial assessment rooms in a mixed department should be capable of receiving patients of all ages.
Staffing and effective workflow	The need for design to allow for staff to have a clear line of sight into waiting room The ability to accommodate an additional Triage Nurse during busy periods The ability to accommodate a Waiting Room Nurse and their equipment requirements Pneumatic tube system in the triage/reception area Two-sided triage with the secondary triage assessment areas to re-assess waiting room patients

Functional content

Spatial relationships

- Access to ECG and diagnostics within the triage area.
- Immediate drug therapy provision (e.g. analgesia).
- Administrative functions like printing
- Patient name-bands or initiating patient tracking at triage along with registration are easily available.
- The need for design to allow for staff to have a clear line of sight into waiting room

Functional requirements	
• Designated areas for ambulance and ambulant patients • It has a single point of entry for acute patients, and provides controlled access to treatment areas and to the wider hospital • Adequate facilities to support dynamic assessment and simultaneous treatment	• Preserve patient privacy and confidentiality • Ensure two exits from each triage cubicle for the safety of assessment nurses • Each triage cubicle equipped with a dedicated alarm • Design considerations to ensure the safety of assessment nurses during patient assessment. • Pneumatic tubes if bloods being taken there.
Specific equipment requirements	
• Examination light • Access to a weighing scale for baby and adults • Mobile patient assessment and monitoring equipment • Equipment and medications for simple treatment measures • Administration of simple treatment measures (e.g. dressing, analgesia, ice, splinting) • Foot stool • Storage for wound dressings, etc.	• Wheelchairs and emergency trolley • Communication system • Service panel • Electronic information entry portal (computer terminal or handheld device) with nearby printer access • Equipment for physiological measurement (vital signs, oxygen saturation, temperature, near-patient blood sugar testing) • Security measures for staff and patients (e.g. duress alarm system and CCTV).

Common design flaws

- ◆ Not having a single triage point of entry for all attendees, ambulance and walk-in patients
- ◆ Not having a triage area located at the entrance hub
- ◆ Wheelchair and disability accessibility not accounted for and how this affects triage desk, administrative office and engagement with clinical staff etc.

Rapid Assessment and Treatment (RAT)

Rapid Assessment and Treatment (RAT) may be offered to expedite care and/or to enhance basic Manchester Triage (MT) in the treatment of minor injuries or illness. It is predominantly staffed by Rapid Assessment Nurses (RANs). Some EDs involve Senior Clinical Decision makers to expedite decision making, investigations and treatment. Various terms, such as Advanced Triage, IMPACT, Team Triage, RASTTA (rapid assessment streaming triage treatment area) and Senior Intervention Following Triage (SIFT), are used interchangeably for RAT in EM.

Size

- ◆ RAT treatment rooms should be sized at 16m²
- ◆ Size of 5m² chair per attendee should be allocated in the sub wait area
- ◆ Total size of area dependent on patient volume and treatment approach.

Design considerations

Sub-wait	Sub-wait areas in the ED should be used to accommodate patients after RAT, rather than the waiting room. Patient experience is adversely affected by moving back to the waiting room. Accessible toilet facilities required, including space for specimen provision.
Emergency assessment area	Some rooms located close to entrances for immediate assessment. Placed immediately after triage, aim to promote forward momentum and early access to diagnostics as well as other major clinical areas Isolation suite at the front door for rapid segregation of potentially infectious patients. Lockable rooms with external emergency override and visual occupancy indicator. At least one room equipped for ENT, ophthalmic and dental problems close by.
Generally	Some rooms close-by, dedicated to a specific function. The area should be fully arranged with gas, suction, power and data points to accommodate any future change in role
Supplies and Equipment	All supplies needed to treat patients in RAT should be held in a supply cart at point of use. Senior clinicians must have appropriate personal protective equipment if undertaking RAT for patients who require isolation.

Functional content

Spatial relationships

- There should be access directly from the reception/triage and waiting room areas without having to pass through other areas
- Easy access to x-ray.

Functional requirements

- The area is for assessment and treatment of patients who are mostly ambulatory and who are anticipated to go home
- The design should be based around a rapid turnover of patients
- Functional space needed here for front-of-house diagnostics e.g., ECG, phlebotomy, cannulation.
- Space required to house equipment such as point-of-care devices, ECG machines Ultrasound Scanner, blood gas analysers, and urinalysis equipment
- Private rooms should be available for communication with ambulatory patients regarding the results of clinical investigations
- Clinical Equipment Store
- Storage space needed for commonly used portable equipment (e.g., ultrasound machine, mobility aids)
- Emphasis on not encroaching on corridor spaces
- Mixture of Chairs and Recliners - ensure all attendees have at least a seat.
- Pneumatic tubes.

Specific equipment requirements

- Have dedicated trolleys (e.g. suture/dressing/cannulation) prepared and stored centrally in the area, rather than stocking each room with all potentially required equipment
- An examination chair or couch
- A service panel
- Wall mounted diagnostic set
- A desk with computer and forms storage
- Sub-wait seating area to cater for volume surge. Adequate space to accommodate wheelchairs
- Essential equipment e.g. ECG machines and supplies must be readily available to optimise RAT efficiency.

Common design flaws

- ♦ Having write up areas that are too small
- ♦ A consultation area that is too small with no ready access to an adjacent waiting area
- ♦ Locating the consulting area away from other clinical areas, resulting in staff feeling isolated
- ♦ Not enough space for supply carts and storage of equipment
- ♦ Not in close proximity to Triage area.

Treatment Area / Majors Area / Urgent Area

A general/acute cubicle is required to assess, manage and initiate treatment on patients with a high likelihood of admission.

Size

Each adult patient treatment area should be at least 16m² in area.

Design considerations

Optimal number of treatment cubicles	Depends on factors related to the overall size of the ED. Estimation involves considering the volume and scope of services provided. Treatment areas should be directly observable from the Staff Duty Base. Special function treatment areas are not included in the calculation for treatment area cubicle numbers.
Physiological monitoring in treatment cubicles	All treatment cubicles should have centrally monitored physiological monitoring.
Privacy for the patient	The cubicle should be able to be closed at the entrance with a sliding partition or curtain Storage for limited amounts of linen and easy access to bedside equipment Direct observation is possible from the staff position
Patient comfort and bespoke needs	A defined number of cubicles fitted with lifting equipment for the disabled or bariatric patient Cubicles catering for the elderly or children
Additional services and dedicated facilities	EDs offering additional services (e.g. forensic examination for victims of sexual violence) require dedicated facilities These facilities should not double as routinely used clinical areas to ensure availability when needed A ceiling rail for IV fluids (though trolleys with IV poles incorporated make transport much easier)

Functional content

Spatial relationships

- Ready access to clean and dirty utility rooms
- Ready access to the medication room
- Ready access to patient toilets/shower.

Functional requirements

- Space to fit a standard mobile trolley
- Space for essential equipment (e.g. oxygen masks)
- Space for accompanying persons (minimum of two people)
- Space to allow procedure trolleys at the bedside, and functional equipment
- Preferably solid walls extending from floor to ceiling for patient privacy
- Ability to close the cubicle entrance with either a sliding partition or a curtain
- Storage for limited amounts of linen and easy access to bedside equipment
- Direct observation possible from the staff position
- Private room for gynaecology/intimate examination.
- Pneumatic tubes.

Specific equipment requirements

- | | |
|--|--|
| <ul style="list-style-type: none">• 2 oxygen outlets• Wall-mounted suction• Service panel with a minimum of ten electrical sockets and four Category 5 sockets• Patient trolley• Patient relative chair• Swivel chair/stool• Handwashing facilities• Surgical light | <ul style="list-style-type: none">• Patient documentation equipment (writing boards/desk)• Computer terminal and diagnostic image viewing facilities• Ophthalmoscope and otoscope• Panic button• Access to dressings and wound care equipment• Sharps bin• Clinical waste disposal bin |
|--|--|

Common design flaws

- ♦ Inadequate footprint resulting in limited circulation space and space for IV trolleys
- ♦ Inadequate lighting and lighting that cannot be dimmed
- ♦ Limited privacy – three sides of cubicle curtained, conversations able to be overheard by adjacent patients
- ♦ A lack of correct disposal bins for infectious and non-infectious waste.

Resuscitation Room(s)

A resuscitation room provides reception, assessment and initiation of treatment for patients who have life threatening or time critical illness. Every Emergency Department should have an adequately sized and formatted Resuscitation Room.

Size

- ◆ Should have a minimum of two Resuscitation bays.
- ◆ Number influenced by acuity, type and volume of cases.
- ◆ A Resuscitation room should be at least 26m² excluding storage space.
- ◆ An additional one quarter to one third of this figure should be allowed for storage.
- ◆ A Non-linear design should be considered to optimise patient monitoring.

Design considerations

Room requirements	The number of resuscitation bays depends on factors such as patient attendance, acuity, case-mix, local clinical practice, services offered, and the need for dedicated Paediatric Resuscitation bays in mixed departments. Trend towards greater use of Resuscitation bays for assessment and stabilization before patients are moved to a less acute setting within ED.
Equipment storage and space allocation	Experience shows deficient equipment storage in Resuscitation Rooms can lead to adverse patient consequences. Some US ED developments now include up to 9m ² for additional equipment storage within the Resuscitation Area. Dedicated area for the storage and charging of transfer equipment.
Access and safety considerations	Easy access between the Ambulance bay and the Resuscitation area. Easy unimpeded access to the resuscitation area from the ambulance entrance, avoiding the main waiting area. Incorporation of radiation protection in the design, including wall lining and staff/patient protection during portable x-ray if individual rooms are not used. Climate-controlled environment and controlled lighting. Access to the body viewing room without passing through public spaces. Maximised visual and auditory privacy for patients, relatives, and staff.
Assembly area and patient reception	Assembly area for the trauma team with the ability to assign roles and gown up. Rapid reception of patients from waiting room, ambulance bay, or helipad. Rapid offload from ambulance stretcher to ED trolley with minimal manoeuvres. Direct access from assessment/treatment rooms for rapid patient transfer. Integration with assessment/treatment rooms; not isolated.

Resuscitation room design	Central mobile trauma trolley with 360-degree access to patients. Medication and equipment stored in or immediately adjacent to the room. Adequate floor space for mobile equipment like ultrasound machines. Adequate shelving for trauma care medication and equipment. Sufficient space to accommodate maximal personnel concurrently involved in treating a patient. An example of excellent resuscitation design is several identical rooms that are all interconnected The ability to maintain patient privacy, whilst allowing ready access and monitoring by clinical staff
Ceiling-mounted fixtures	Recommendation for ceiling-mounted rather than floor-mounted pendants. Powered ceiling-mounted pendants for easy operation by staff of all heights. Careful positioning of pendants for convenient staff access. Each pendant connected to an isolated power supply with an uninterruptable power supply (UPS). Preference for ceiling-mounted X-ray systems; radiation protection considerations. Coordination of design for all ceiling-mounted equipment and lighting to avoid compromise in use

Functional content

Spatial relationships

- Access from waiting room, ambulance bay, or helipad
- CT scanner close to the resuscitation room
- Ready access to radiology department and operating theatre/Intensive Care Unit
- Desirable for resuscitation rooms to be equipped for radiology
- Ultrasound machine for resuscitation scanning
- Hand-held Arterial Doppler machine available
- Viewing Room for family member viewing of very recently deceased patients whose resuscitation in the ED has been unsuccessful prior to their transfer to the hospital Mortuary
- Adjacency to distressed relatives' room, providing privacy in transit to the resuscitation rooms
- Near-patient testing facility, including blood glucose, Arterial Blood Gas measurement, and other tests.

Functional requirements

- Sufficient space for 360-degree access to the patient on a high-specification resuscitation trolley suitable for portable x-ray acquisition
- Full individual physiological monitoring, including ECG, NIBP, oxygen saturation, core temperature, invasive monitoring, end-tidal CO2 monitoring, etc.
- One ventilator for each resuscitation bay.
- Three oxygen outlets.
- Wall-mounted suction.
- Resuscitation equipment for intubation, defibrillation, with external pacing capabilities, cannulation, etc.

- Equipment or overhead rail for IV fluids.
- Handwashing facilities and clinical wash basin
- Clock with timing facilities, synchronised against an accurate time source.
- Operating theatre quality light
- Wall-mounted ophthalmoscope/otoscope
- Overhead X-ray gantry or alternative immediately accessible imaging system.
- Clinical waste facilities for sharps and clinical waste
- Dedicated IT terminal with access to Laboratory and Diagnostic Imaging results and the ability to request clinical investigations.
- Staff call panel
- Pneumatic tubes.
- Physiologic monitor networking plus remote monitor
- Temperature control for patients undergoing surgical procedures.
- Consideration of equipment and fixtures for efficient and safe care, such as:
 - » Ceiling-mounted twin-armed pendant.
 - » Patient trolley and space for dressings and instruments trolleys.
- “Grab” board for small items storage.
- Whiteboard
- Sack holders for clinical waste and general waste
- Boards for recording progress, drugs, etc.
- Screens for posting algorithms.
- Storage space for sterile and non-sterile consumables.
- Outlets and connections on each pendant for flexibility:
 - » 14 x 13-amp socket outlets (HBN)
 - » Data outlets, networked to the hospital's patient record system.
 - » Oxygen and medical vacuum outlets.
 - » Anaesthetic gas scavenging points.
 - » Call systems and telephone outlets.
 - » Dimmer switch for overhead lighting
- Equipment considerations for connection to each pendant:
 - » Hands-free telephone.
 - » Multi-parameter patient monitoring equipment.
 - » Volumetric pumps, syringe drivers, blood warmer.
 - » Transport ventilator.

Specific equipment requirements

- Resuscitation equipment sets for advanced airway management, surgical airway, emergency surgical procedures, urinary catheterization, central line placement, etc.
- Two transport monitors and ventilators for patient transfer.
- Facilities for non-invasive ventilation
- Twin diagnostic imaging monitors.
- Standard Anaesthetic Machine.
- Medical air and nitrous oxide.
- Locked cupboard for controlled drugs
- Facilities for warming resuscitation fluids.
- Area for patient documentation / note-keeping / IT system access and ability for nurse to act as a scribe.
- Sterile procedure preparation area.
- Access to O-negative blood for urgent transfusion.
- Resuscitation equipment storage with re-charging capabilities
- Utility area.
- Paediatric resuscitation equipment, including a Resuscitaire for Neonatal Resuscitation and warming equipment.
- Refrigerator
- Adequate number of intravenous pump devices

- Equipment to warm patients and create an appropriately warm environment for the assessment of patients with major burns.
- Adequate worktop space for PCs, monitors, documentation etc.
- Rapid infusion device

Common design flaws

- ◆ Inadequate space to accommodate the maximum staff personnel concurrently involved in treating a patient
- ◆ Resuscitation areas that are located too far from the ambulance bay
- ◆ CT scanner too far from the resuscitation area
- ◆ Not using a standardised design configuration

Isolation room

A negative/positive pressure isolation room in an ED is designed to prevent the spread of airborne infectious diseases. It ensures that air flows into the room and not out, reducing the risk of contaminants escaping. This specialised isolation space is crucial for safely managing patients with contagious respiratory illnesses, helping to protect healthcare staff and other patients.

Size

- ◆ Every ED should have at least two isolation rooms for direct transfer and assessment from ambulance or triage.
- ◆ At least one room should be designed for children and another suitable for adult patients.
- ◆ Each room should be 16m² i.e. same as a treatment bay.

Terminology

- ◆ Health Building Note 04-01
- ◆ Class S – Positive pressure: Patient protection – Type 4
- ◆ Class P – Positive pressure: Patient protection – Type 3
- ◆ Class N – Negative pressure: Respiratory isolation – Type 5

Design considerations

Room design and flexibility	At least one room should be designed for children and suitable for adult patients. Each room should be of similar size to an acute treatment bay to allow for the treatment and resuscitation of critically ill isolated patients if required
Supplies and location	The ante-room should be of sufficient size to allow for adequate supplies of PPE, linen and supplies and for waste disposal inside the room Isolation rooms should be located near outer walls to facilitate ventilation requirements.
Infection control	Infection control staff should contribute to the needs assessment, design and operational policy for isolation rooms and the entire ED build. Isolation rooms should be equipped with physiological monitoring linked to the centrally monitored ED system.

Patient-specific isolation requirements	Type 5 negative pressure rooms are not suitable for potentially immunocompromised patients; these patients require a Type 4 standard isolation room or Type 3 positive pressure patient protection room. Isolation rooms may also be used to treat patients with conditions that require separation from other patients, such as those needing privacy for clinical conditions or who are a source of visual or auditory distress to others.
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Functional content

Spatial relationships

- Positioning of these rooms should be adjacent to the triage area where patients are received to allow for the immediate isolation of potentially highly infectious patients without the need for the patient going through and potentially contaminating or infecting other patients in the ED.
- Each ED should have one Type 5 isolation room with additional requirements being determined by hospital location, role and patient demographics.

Functional requirements

- Respiratory isolation rooms for patients who require airborne droplet nuclei isolation (Type 5) should have negative ventilation, an ante-room with scrub up facilities (Type A hand-basin), and be self-contained such that they have shower/toilet en-suite facilities that are compliant with HBN standards
- The air handling systems designed for airborne infection isolation should be connected to the emergency backup power in case of power failure.
- A communication system should be provided so that staff and patients can communicate with people outside the room without having to leave the room.

Specific equipment requirements

- The room should be fitted as per acute treatment areas
- Macerators for treatment of human waste or individual equipment sterilisers should be considered
- Type A clinical basin – large (scrub basin)
- Isolation rooms should be equipped with en-suite toilets.

Common design flaws

- ♦ Not working with infection control / public health teams on all stages of design from planning through to delivery of Isolation Rooms
- ♦ A Type 5 isolation room that cannot be securely locked from the inside if it has a separate entry in addition to the ante-room. Such a room must not be able to prevent entry whilst in use as a Type 5 isolation room
- ♦ Not having enough negative pressure/positive pressure rooms for the demographic type of patients attending the hospital

Diagnostic imaging

Diagnostic imaging (DI) in an ED serves to swiftly and accurately assess injuries or illnesses, aiding in rapid diagnosis and treatment decisions using Emergency Radiology modalities. It helps identify critical conditions, guide interventions and streamline patient management for better outcomes. The goal is to provide timely and precise information to support emergency care and improve patient outcomes.

Size

The size of a DI department will be determined by the volume and type of presentations and there must be close collaboration with both the DI and ED teams to optimise speed, workflow, and pathways. Minimum acceptable floor area for the following areas are:

- ♦ General X-Ray Imaging Room - 33m²
- ♦ CT Scan/MR Imaging Room - 44m²
- ♦ Ultrasound Imaging Room - 16m²
- ♦ Imaging viewing and reporting - 12m²
- ♦ Shared control room - 20m²

Design considerations

Requirements	Dedicated 24/7 access to both digital plain imaging and CT scanning is essential. Provision of facilities as part of the ED development or in close proximity. Consideration of access to MRI and Interventional Radiology in larger centres. Close proximity to resuscitation areas, with consideration given to integrating CT into one of the Resus Bays.
Other considerations	Plain radiology plays a significant role in supporting ED. Heavily used in the diagnosis of suspected fractures or other forms of trauma. Refer to the Trauma Imaging section for additional details of HBN imaging notes <i>Facilities for diagnostic imaging and interventional radiology (england.nhs.uk)</i>

Functional content

Spatial relationships

- Access to emergency diagnostic imaging for all patients and, in particular, stroke and trauma patients. Increasingly ongoing resuscitation takes place in parallel to CT scanning the patient.
- Reporting office for radiology
- Area for radiographer's processing
- Sub-wait area for ambulatory patients to wait +/- drink contrast

Functional requirements

- Implementation of an alarm system for prompt response
- Means of communication with the ED duty base
- Inclusion of a PACS viewing station in every patient treatment cubicle
- Provision of at least three twin monitor viewing systems at the staff duty base
- Ensuring ease of access for medical and rANP staff to the viewing systems.

Specific equipment requirements

- For further information check *Facilities for diagnostic imaging and interventional radiology (england.nhs.uk)* doc

Common design flaws

- ♦ Lack of proximate flow with ED
- ♦ Limited flexibility in room configurations
- ♦ Inability to move portable scanning equipment seamlessly from DI department to ED
- ♦ Dispersed office areas hindering effective communication
- ♦ Inadequate space planning hindering potential growth
- ♦ Limited flexibility in room configurations.

Mental health consulting rooms

EDs should be designed for the reception, assessment, and initial treatment of individuals with acute mental health issues. They are not meant to replicate dedicated mental health admission centres but rather focus on immediate assessment and care. The best practice guidance for mental health from the Health Service Executive (HSE), Royal Colleges of Emergency Medicine and the Colleges of Psychiatry are essential resources for ensuring quality care in mental health emergencies and all designs should align.

Size

- ♦ The minimum acceptable floor area per consultation room is 16m².
- ♦ Ideally, the space should contain at least two separate but adjacent areas including an interview room and examination room.

Design considerations

Integration with ED operations	Acute presentations of patients with behavioural disturbances can disrupt the usual operation of an ED. The busy ED environment may not be conducive to the care of patients with acute mental health illness.
Medical co-existence	Patients with acute mental health crises may have concurrent medical problems. Life-threatening illness or injury remains the first priority and should be managed within the appropriate clinical area of the ED.
Patient supervision	Uncontrolled patients should never be left unsupervised in any ED area. Acute mental health area should be remote from paediatric areas.
Patient safety	Primary purpose is to provide a safe and suitable space for patient interviews and stabilization. All window furnishings such as shading devices and others, should be appropriately designed and located so that they cannot be accessed by patients and potentially be used for self-harm. The area should be free of mobile or breakable furniture, sharp or hard surfaces which could injure an uncontrolled patient – furniture made mostly of foam rubber has an advantage in this regards. CCTV in any mental health assessment areas should be concealed in order to minimise distress for those patients suffering from paranoia. The area should be arranged to ensure that patients have no access to air vents or hanging points, and should also incorporate tamper resistant electrical fittings. The area should be shielded from external noise and designed in such a way that direct observation of the patient by staff outside the room is possible at all times. Medical gasses, if fitted, should be located behind a solid non tamper proof roller panel accessible by staff via swipe card access. Ideally, the area would have separate entrance and exit points.

Safety for staff	The room should not contain objects that could be thrown at staff. The exit doors should open outwards. If a window is incorporated, it should be made from shatter-proof material.
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Functional content

Spatial relationships	
The acute mental health assessment facility should ideally be located in, or adjacent to, the ED.	A separate mental health facility needs to be close enough to the main staff base area of the ED to allow for adequate response and observation, unless it is to have its own staffing.
Functional requirements	
- Provide an area that is conducive to the assessment of patients with mental health issues and for discussion with family/friends - Ensure patient and staff safety - Décor promoting a sense of calm with consideration of patient comfort	- Easy access to assistance should a threatening situation arise - Room should be decorated in calm colours - Well-lit with non-breakable glass lighting - Should have an observation window - Viewable from outside.
Specific equipment requirements	
- Smoke detector - Duress alarm - Electrical outlets and medical gasses (if installed) must be inaccessible to patients	- Panic button 2 exit routes - Fixed furniture and fittings that cannot be used as weapons

Common design flaws

- ♦ Designing a self-contained area that cannot be adequately staffed or monitored creates clinical risk and becomes useable.
- ♦ This results in patients being managed in a general ED environment that does not meet the needs of this patient group
- ♦ Prolonged observation of uncontrolled patients is not the responsibility of EDs.

Eye Casualty / Eye Room

An eye room in an ED serves to address urgent eye-related concerns and injuries promptly. It provides specialised care for a range of ocular emergencies, including trauma, infections, and other acute conditions. The goal is to offer timely assessment, treatment, and, if necessary, referral to the local Ophthalmology service. Combining specialty rooms such as the Eye Room, ENT Room, and Plaster Room into a single, multi-use space is common practice, maximizing space efficiency and flexibility and significantly enhancing the functionality and adaptability of clinical areas.

The eye room should be at least 12m² (unless combined with other rooms e.g. ENT room). However, it is advisable to consider room size of 16m² due to equipment and storage.

Design considerations

Generally	Combining the eye room with ENT room Place slit lamp lighting on timer Wall mounted ophthalmoscope and otoscope for those patients not tolerating slit lamp examination Ability to dim room lights rather than an on/off light switch
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Functional content

Spatial relationships	
To be near the ambulatory / discharge / Fast Track area of an ED as most patients stay a short period and will go home after treatment.	- Should be located near to wait and triage area - If positioned outside of the ED, needs to be a core adjacency
Functional requirements	
- Consultation room type set up - Slit lamp examination facility - Ability to dim lighting in room	- Ability for patients to sit reclined - Ability to irrigate a patient's eye - Ability to black out the room.
Specific equipment requirements	
- Desk with storage area for forms - Diagnostic chart - Eye chart – traditionally a 6-metre chart. Consider using a 3-metre chart (eliminates the need for reflective mirror) which is best if well illuminated - Marking on wall/floor to determine correct distance from eye chart - Sink / basin / facility to irrigate eye - Slit lamp and two examination chairs	- Ability for patient to lie recumbent either in chair or on examination couch - Shelving for eye equipment (eye pads/shields/cotton buds) - Refrigerator for eye medication requiring refrigeration - IT desk to enable patient to be seen, treated and discharged from room.

Common design flaws

- ♦ A lack of circulation space around the slit lamp
- ♦ Inability to lie patient down, if required
- ♦ Not having access to both sides of the patient
- ♦ Having the eye room as part of another room, rather than it being a separate room.

Ear, Nose and Throat room

The Ear, Nose and Throat (ENT) room is used for the examination and treatment of ENT conditions, such as removal of foreign bodies in the ear or epistaxis. Combining the three specialty rooms of the Eye Clinic, ENT Room, and Plaster Room into a single, multi-use space is common practice, maximizing space efficiency and flexibility and significantly enhancing the functionality and adaptability of clinical areas.

Size

The ENT room should be at least 12m² (unless combined with other rooms e.g. eye room). However, it is advisable to consider room size of 16m² due to equipment and storage.

Design considerations

Generally	The room serves for the examination and treatment of various ENT conditions, including procedures like removing foreign bodies in the nose ear or managing epistaxis. Combining the ENT room with the ophthalmology room allows for a consolidated approach to addressing head and neck-related concerns efficiently. The design should include secure storage for therapeutic medications such as otitis externa ointments/drops and silver nitrate sticks, ensuring easy access and adherence to security protocols in the ED.
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Functional content

Spatial relationships	
To be in the ambulatory / discharge / Fast Track area of the ED, as most patients stay a short period and will go home after treatment	The dirty utility room is for the disposal of blood-stained linen.
Functional requirements	
- Adequate size to allow a trolley - Appropriate physiological monitoring systems and resuscitation equipment to allow safe analgesia and sedation - Access to the patient from both sides	Adequate room/storage capacity to allow for procedure trolleys and equipment.
Specific equipment requirements	
- Staff desk with computer, forms storage - Diagnostic set on wall - Chair/couch able to position the patient anywhere between supine and upright - ENT microscope - Wall mounted suction - Oxygen, suction, monitoring equipment	- Shelving or ENT trolley containing necessary equipment for treatment of most common ENT presentations e.g. ear specula, nasal packs, suction tubing - Head lamp storage - Ready access to clean and dirty linen - Infectious and non-infectious waste disposal.

Common design flaws

- ◆ A lack of circulation space and access to both sides of the patient
- ◆ Inability to lie the patient down if required
- ◆ Inability to manage a patient who collapses
- ◆ Having the ENT room as part of another room, rather than it being a separate room

Plaster room

The plaster room is used not only for the application of splints, plasters and mobility aids, associated with musculoskeletal injury, such as dislocations and fractures but also joint and fracture reduction using procedural sedation.

Combining the specialty rooms of the Eye Clinic, ENT Room, and Plaster Room into a single, multi-use space may be considered, maximizing space efficiency and flexibility and significantly enhancing functionality, storage and adaptability of clinical areas.

Size

The room should be at least 16m² in size. If combined with other rooms, such as a procedure room or procedural sedation room, it is advisable to have two. However, considering equipment and storage needs, it is advisable to aim for 20m².

Design considerations

Generally	There should be enough floor space to accommodate an image intensifier or portable x-ray machine, when required The room should have dimmable lights Special sink facilities needed for plaster material Storage of pre-formed Orthopaedic Splints which requires considerable space Designated area close to ambulatory care for storage recommended Plaster of Paris and splinting equipment in Resuscitation room may also need storage in the designated area
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Functional content

Spatial relationships	
- The plaster room is in the ambulatory / discharge / RAT area of the ED, as most patients stay in the plaster room a short time and will go home, to a ward or elsewhere in the ED after treatment - Adjacency to Diagnostic Imaging and Resus	- There should be consideration of placing the plaster room close to both minor and major areas of the ED if sedation is likely to occur in the room - The close proximity of the plaster room to the medication area would allow for ready access to local anaesthetics.
Functional requirements	
- Adequate size to allow a trolley - Appropriate physiological monitoring systems and resuscitation equipment to allow safe analgesia and sedation - Access to the patient from both sides	- Adequate room/storage capacity to allow for procedure trolleys and equipment e.g. tourniquet matching - Room to accommodate portable x-ray or image intensifier.
Specific equipment requirements	
- Staff workstation with computer, phone and form storage - PACS viewing on dedicated PACS terminal	- Fixed or portable nitrous oxide delivery system - Operating theatre light suspended from the ceiling with minimum 80,000 lux - Plaster equipment and plaster trap

- | | |
|---|--|
| <ul style="list-style-type: none">• X-ray viewing box for printed films (usually bought in by patient from external source)• Monitoring equipment and access to resuscitation equipment• Services panel | <ul style="list-style-type: none">• Plaster saw with vacuum fitting• Other storage for splints, braces, crutches• Pneumatic tubes.• Dimmable lights for ketamine administration |
|---|--|

Common design flaws

- ♦ A plaster room that is cluttered, reducing access to both sides of a patient.
- ♦ Storage requirements are not adequately addressed.
- ♦ Not having a separate plaster room.

Procedure room

The procedure room(s) may be required for the performance of procedures such as suturing, phlebotomy in very young children, lumbar puncture, tube thoracostomy, thoracocentesis, abdominal paracentesis and bladder catheterisation.

Size

The procedure room should be at least 24m² in size (unless combined with other rooms e.g. plaster room). For larger EDs, it is advisable to have two procedure rooms.

Design considerations

Paediatric patients	The room should be furnished to accommodate for paediatric patients
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Functional content

Spatial relationships	
To be in an ambulatory / discharge / Fast Track area of the ED as most patients stay in the procedure room for a short time and will go home or to a ward or elsewhere in the ED after treatment	Consideration is required as to the optimal location and proximity to the resuscitation room, if conscious sedation is likely to occur frequently.
Functional requirements	
- Adequate size to allow a trolley - Soundproof (from external auditory stimuli). - Adequate room / storage capacity for various procedure trolleys, including sutures, dressings and resuscitation equipment.	Designed to facilitate advanced airway management and resuscitation in case of patient deterioration during a procedure
Specific equipment requirements	
- A computer screen for digital imaging viewing and use as a workstation - Monitoring equipment including access to resuscitation equipment - Wall oxygen and suction - Nitrous oxide delivery system or storage space for a portable nitrous oxide delivery system	- Storage for suture equipment, dressings, catheters, sterile packs/procedure bundles - Image intensifier and/or x-ray machine if room is to be used for orthopaedic procedures - Operating theatre light suspended from the ceiling with minimum 80,000 lux - A procedure room should have dimmable lights.

Common design flaws

- ♦ Inadequate floor space that does not allow for ready access to both sides of a patient.

Procedural Sedation and Analgesia Room

A procedural sedation room in an ED is a specialized area designed for carrying out procedures that require sedation and pain management.

Size

The procedure room should be at least 24m² in size (unless combined with other rooms e.g. plaster room).

Design considerations

Paediatric patients	The room should be furnished to accommodate paediatric patients
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Functional content

Spatial relationships	
located close to the radiology hub and in proximity to resuscitation areas for timely escalation of care	Consideration is required regarding placing the procedure room close to both minor and major areas of the ED, if conscious sedation is likely to be used.
Functional requirements	
- Appropriate physiological monitoring systems to allow safe analgesia / sedation - Soundproof (from external auditory stimuli). - Adequate room / storage capacity for various procedure trolleys, dressings and resuscitation equipment.	- Designed to facilitate advanced airway management and resuscitation in case of patient deterioration during a procedure - Nitrous oxide provision (wall-mounted with scavenging) - A low-stimulation environment are recommended, with dimmable lights suggested
Specific equipment requirements	
- A computer screen for digital imaging viewing and use as a work station - Monitoring equipment including access to resuscitation equipment - Wall oxygen and suction - Nitrous oxide delivery system or storage space for a portable nitrous oxide delivery system	- Storage for procedure trolleys, suture equipment, dressings, catheters, sterile packs/procedure bundles - Image intensifier and/or x-ray machine if room is to be used for orthopaedic procedures - Operating theatre light suspended from the ceiling with minimum 80,000 lux - A procedure room should have dimmable lights.

Common design flaws

- ♦ Inadequate floor space that does not allow for ready access to both sides of a patient.
- ♦ Placed far from Resus and critical areas

Multi-disciplinary patient assessment areas

Designated space for patient assessment by Occupational Therapists (OTs) and Physiotherapists may be required within the ED to allow mobility assessments and falls prevention interventions to be undertaken. Close relationship with a Geriatric Emergency Medicine (GEM) Unit, if available on site, and may be co-located or adjacent to it.

Size

Advisory to have a large multi-use space of 32m² that can be split into two 16m² MDT rooms or one large room.

Size will vary depending on the size of the ED and the allied health professionals' caseload. If relevant, this may be appropriately combined with a CDU or MAU/SSU/GEM type facilities.

Design considerations

Element	Description
	This work is especially important to support the safe ED discharge of elderly patients. Facilities for multi-disciplinary care are critical in order to allow medical, nursing and allied health staff to work together effectively to assess and manage patients in the ED. Frailty Liaison Clinic in EDs https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9819173/

Functional content

Spatial relationships	
Ideally to be located in the ED or, at least, adjacent to it.	An OT/physio assessment area, if included, should be adjacent to MAU/SSU or ED.
Functional requirements	
- An area with computer access and ability to hold confidential discussions or phone calls - An OT/physio assessment area - Will need to be equipped for staff such as Specialist Falls rANP, dietician, social worker, Speech and Language Therapist	- Storage for equipment that is loaned to patients. - Will require plinth, steps etc. for assessment by physiotherapy
Specific equipment requirements	
Mats, physio equipment	OT equipment

Common design flaws

- ◆ Not close to ED
- ◆ Located near Mental Health Consultation Rooms
- ◆ Positioned on floors other than the ground floor
- ◆ Accessibility for patients hindered by steps or stairs.

Paediatric Emergency Medicine facilities in mixed EDs

Designing facilities within Mixed Emergency Departments that cater to the unique needs of young children accompanied by their families or caregivers is an important consideration. The Children's ED should be audio-visually separated from the adult ED.

Size

- ♦ Floor space allocation dependent on overall ED size and paediatric patient volume
- ♦ Minimum acceptable floor areas: consulting room - 16m²; trolley area - 12m².

Design considerations

Element	Description
	Dedicated space for assessing paediatric patients with mental health conditions. Inclusion of television/video facilities.

Functional content

Spatial relationships	
• Ensure visibility of the paediatric waiting area to nursing staff at all times.	
Functional requirements	
• Dedicated waiting area with acoustic and visual separation from main waiting area • Appropriately decorated paediatric waiting room with a play area • Adequate space for prams, parents, and siblings • Resuscitation room • Treatment rooms accessible from the paediatric waiting area	• Appropriate decor for different age groups • Trolley-based treatment room • Consideration for placing the resuscitation room alongside or within other resuscitation facilities • Incorporation of low handles in rooms accessible to children, and high handles with self-closing doors in restricted areas (e.g., laundry, medication preparation).
Specific equipment requirements	
• Weigh scales • Infant Resuscitaire • Equipped similarly to adult rooms with age-appropriate medical equipment • Consider a second medical services panel dedicated to paediatric patients • Provision of paediatric-sized toilets	• Seating for both adults and children • Designated area for baby change and breastfeeding • Adequate space for prams and wheelchairs • Dedicated space for assessing paediatric patients with mental health conditions • Television/video facilities
Standalone Paediatric emergency medicine facilities requirements	
• Paediatric resuscitation • Paediatric Procedure Rooms • Distressed Parent's/Relatives Room • Breastfeeding Room	• Sensory room • Isolation Room, especially for unvaccinated infants under 8 weeks olds

Common design flaws

Avoid designing small rooms for children that limit their use exclusively for paediatric purposes.

Clinical decision unit (CDU)

Clinical Decision Unit caters for short-term treatment, observation, assessment, and reassessment of a broad group of patients including the vulnerable (Overdose, Deliberate Self Harm, Detoxification, 1st seizure). A Paediatric CDU may also be considered along with an Adult CDU as two separate spaces.

Size

The goal is to incorporate individual spaces to ensure comprehensive infection control measures.

- ♦ Minimum area per bed space: 16m²
- ♦ Minimum acceptable floor areas: CDU room - 16m²; CDU bed space - 12m²

Design considerations

Admission and discharge policies	Specific admission and discharge criteria and policies. Establish clear hospital-wide patient pathways, including referral and diagnostic protocols, alongside ensuring safe staffing levels, both clinical and non-clinical. CDU should be used for short stays only with length of stay always targeted at less than 24 hours, physically separated from the ED's acute assessment area.
Facility design and resources	Static number of beds with oxygen, suction, and patient washing facilities Ability to monitor each bed at the same level as an acute cubicle with dedicated staff bases. Utilisation of hospital beds, not ED trolleys.

Functional content

Spatial relationships

- Proximity to the ED for efficient collaboration between EM staff.
- Bed capacity depends on in-patient services scope and projected length of stay.
- A Resuscitation Trolley and Defibrillator must be easily accessible for all beds.
- Otoscopes and ophthalmoscopes must be available.
- Emphasis that ED in-patient care areas should not be part of the general hospital bed-stock
- Unacceptable for inpatient boarders to be accommodated in this area

Functional requirements may include

- Observation Ward
- ED communications system coverage for the Observation ward; alarm system for emergency calls.
- Observation ward requirements:
 - » standard ward facilities
 - » relatives/interview room
 - » duty station
 - » dirty utility
 - » drug preparation and storage area

- Hospital beds rather than ED trolleys recommended for increased patient comfort.
- » Patient showering and toilet facilities.

Specific equipment requirements

- An oxygen outlet
- Wall mounted suction
- A service panel with a minimum of ten electrical sockets and four category 5 sockets.
- Patient relative chair
- Hand-washing facilities
- Patient locker
- Nurse call system
- A panic button.

Common design flaws

- ♦ Inability to monitor each bed at the same level as an acute cubicle
- ♦ No dedicated staff bases.

Dedicated infusion preparation room

A pharmacy/medication room is used for storage and dispensing of medications, and the preparation of drugs (including IV administered). Its function may extend to clean utility, medication storeroom.

Essential for managing the safe delivery of pharmaceutical products.

Medication dispensing solutions like Pyxis are available that help secure tracking of medications to patients, stock take etc. These have a space requirement.

Size

The Dedicated Infusion Preparation Room should be a minimum 16m².

Design considerations

Medication storage and security	Appropriate pharmacy storage systems. Adequate workspaces for drug preparation. Disposal facilities for proper waste management. Security swipe card-controlled access for controlled entry. Drug Storage to conform to national standards for drug storage areas.
Equipment and infrastructure	Space for the storage and charging of IV infusion pumps. Cold water dispenser (e.g. hydro tap). Refrigerator monitor and alarm (linked to Pharmacy and Security) to prevent warming of temperature-sensitive medications. Consideration for space to accommodate a medication trolley. Wall-mounted push-button alert indicating to staff outside the pharmacy/medication room that the controlled drug cupboard keys are required. Consider space requirements and inclusion of a Pyxis MedStation or electronic pharmacy management/dispensing system
Automated dispensing considerations	If an automated dispensing machine is installed, consider omitting swipe card door access in favour of open access to the room with secure storage of medication.

Functional content

Spatial relationships

- The pharmacy / medication room should be situated to allow easy access to all clinical areas.

Functional requirements

- Secure electronic medication storage solution, and open access to pharmacy / medication room (preferable) or
- Storage space and shelving for all ED medication.
- Centralised - single medication area serving all of an ED or
- Decentralised - multiple medication areas, more common in larger EDs or those with electronic medication storage solutions.

Specific equipment requirements

- Shelving for syringes, labels, medication containers and other equipment required for the preparation of medication
- Secure electronic medication storage including power and data points or
- Storage space and shelving for all ED medication including:
 - » Shelving for storage of medication;
 - » Lockable cupboard for restricted medications;
 - » Cupboard for take home and after hours medication (pre-packs);
- Secure lockable cabinet secured to the wall for restricted schedule drugs.
- Consider digital keypad access to control drugs
- Shelving for IV fluid used to prepare infusions
- Secure box for the placement of medication that is to be returned to the pharmacy
- Shelving for written resources not available electronically
- Sharps container(s)
- Non-infectious waste bin
- Sink(s)
- Computer (for references, guidelines and electronic medication charts).

Common design flaws

- ♦ Not having sturdy shelving or baskets to support IV fluids

Clean utility / medication room

The Clean Utility/Medication Room in the ED is a sterile space for storing and preparing medications and medical supplies, ensuring rapid access, accurate administration, and adherence to safety protocols. It may be co-located/consolidated with an infusion preparation room.

Size

The pharmacy/medication room/clean utility room should be a minimum 16m².

Design considerations

Pharmacy and medication room design	The pharmacy/medication room/clean utility room should be a minimum of 16m². Space for the storage and charging of IV infusion pumps. Cold water dispenser (e.g., hydro tap) for pharmaceutical use. Refrigerator monitor and alarm (linked to security) to prevent warming of temperature-sensitive medications. Consideration of space for a medication trolley. Consider space requirements and inclusion of a Pyxis MedStation or electronic pharmacy management/dispensing system (of around 2m²)
Security and access measures	Wall-mounted push-button alert indicating to staff outside the pharmacy/medication room that controlled drug cupboard keys are required. If an automated dispensing machine is installed, consider omitting swipe card door access in favour of open access to the room with secure storage of medication.

Functional content

Spatial relationships	
The clean utility/medication room needs to be situated so as to allow easy access to all clinical areas	
Functional requirements	
- Secure electronic medication storage solution, and open access to pharmacy/medication room (preferable); or - Storage space and shelving for all ED medication, including Imprest system.	- Centralised - single medication area serving all of an ED; or - Decentralised - multiple medication areas, more common in larger EDs or those with electronic medication storage solutions.
Specific equipment requirements	
- Secure electronic medication storage including power and data points or - Storage space and shelving for all ED medication including: » Shelving for storage of medication » Lockable cupboard for restricted medications » Cupboard for take home and after-hours medication (pre-packs) » Secure steel lockable cabinet secured to the wall for restricted schedule drugs » Consider digital keypad access to control drugs	- Shelving for syringes, labels, medication containers and other equipment required for the preparation of medication; - Shelving for IV fluid used to prepare infusions; - Secure box for the placement of medication that is to be returned to the pharmacy; - Shelving for written resources not available electronically; - Sharps container(s); - Non-infectious waste bin; - Sink(s); and - Computer (for references, guidelines and electronic medication charts).

Common design flaws

- ♦ Not having sturdy shelving or baskets to support IV fluids

Dirty utility area

The dirty utility/disposal room is used for the disposal of clinical and other wastes and soiled linen; for testing and disposal of patient specimens; for decontamination and storage of patient utensils such as pans, urinals and bowls and for cleaning and holding of used equipment for collection and sterilisation elsewhere.

Size

The minimum acceptable floor area for a Dirty Utility Area is 8m².

Design considerations

Designated dirty utility areas	Ensuring designated areas for the disposal of waste materials. Easy access from all clinical areas for efficient waste management.
Continuous operation and cleaning	Recognizing the 24/7 operational nature of EDs. Establishing facilities for ongoing cleaning to maintain hygiene standards.
Storage for cleaning materials	Allocating dedicated storage spaces for cleaning materials within the ED. Ensuring accessibility for staff to readily access and utilise cleaning supplies.
Other considerations	A macerator for disposable bedpans/urinals rather than washer/sanitiser Two or more dirty utility rooms may be required to minimise travel distance in a large ED. Consider single patient pan cleaners and macerators in isolation room en-suites

Functional content

Spatial relationships	
Access should be available to this room from all clinical areas	
Functional requirements	
There must be a sink for hand washing and a rim sink that is able to be flushed	There should be sufficient space to house the equipment listed below
Specific equipment requirements	
- Stainless steel bench top with sink and drainer - Pan and bottle/urinal rack - Bowl and basin rack - Utensil washer - Pan/bowl/utensil washer/sanitiser	- Flushing sink - Storage space for testing equipment e.g. urinalysis - An optional disposal room adjacent to the dirty utility should be considered.
Other considerations	
- A macerator for disposable bedpans / urinals rather than washer / sanitiser - Consider single patient pan cleaners and macerators in isolation room en-suites.	Two or more dirty utility rooms may be required to minimise travel distance in a large ED.

Common design flaws

- ◆ Not having adequate provision for dirty utilities in treatment zones. Dedicated dirty utilities needed in generating zones.
- ◆ A pan room/area where urine testing takes place that is not separate from the storage of clean equipment.

Decontamination area

A decontamination room is used to decontaminate patients prior to entering the ED, including those who have potentially or actually been exposed to or contaminated with toxins, chemicals, radioactive materials and other hazardous substances. This area will accommodate both ambulatory and non-ambulatory patients.

Size

- ◆ The Decontamination / Patient Cleansing / Wet Room must have a minimum area of 16m²
- ◆ The Decontamination Equipment Store must have a minimum area of 4m², however would ideally be 8m² or more.
- ◆ The size of such a decontamination room is dependent on a hospital's role in the regional response to Mass Casualty Incidents.

Design considerations

Preventing contamination in ED facilities	Avoiding contamination in the event of chemical, biological, or radiation hazards. Constantly adapting to evolving best practice guidelines for decontamination.
Incorporating international standards	Applying current international standards during the design phase of the ED. Regularly reviewing facilities to ensure ongoing compliance and appropriateness.
Decontamination facility options	Evaluating the use of external facilities (e.g., inflatable units) or internal decontamination. Ensuring the availability of means and appropriately trained personnel for assembling external equipment during major incidents.
Service and storage requirements	Establishing dedicated services for decontamination facilities within the ED. Allocating storage areas for decontamination equipment.
Engineering specifications for water drainage	Implementing and maintaining a separate drainage system for decontamination to avoid integration with the main hospital drainage system.
Other considerations	Ideally, tepid water should be available via the water hose/shower In planning this room, there is a need to think through a decontamination pathway – external access separate from normal access Engagement with local fire services may also be required in terms of planning and space required for decontamination areas Consideration of contaminated water collection tank depending on local regulations

Functional content

Spatial relationships	
The decontamination room should be directly accessible from the ambulance bay without entering any other part of the ED.	
Functional requirements	
- A flexible water hose, floor drain and contaminated water trap (if required by regulation) - Both ventilation and drainage systems must be independent and capable of being isolated - Privacy from public view - Area to perform immediate resuscitative care by clinical staff	- Ideally, there is separation of ambulant and non-ambulant patients - A decontamination area consists of: » de-robing area » decontamination area including water hose » drying off area » entry to the ED proper.
Specific equipment requirements	
Demountable shower	Benching/racks

Common design flaws

- ♦ Not located near to Ambulance Access.

Staff base

Staff work stations are where staff can undertake work that is not performed at a patient's bedside. They are used for discussion, for advice and referrals, and for the entry or writing of notes. There is a need for a range of private meeting rooms that will enable confidential discussion and collaboration between staff. Separate consulting rooms will be provided specifically for patient consultation.

Size

- ◆ The staff duty base should be a minimum of 12m².
- ◆ Alternatively, 5.5sqm per person
- ◆ The staff base should serve to oversee a team of 6-8 staff.

Design considerations

Safety and confidentiality	Ensuring the area promotes staff safety and patient confidentiality. Enclosing part of the workstation to increase privacy. Need to allow staff to perform appropriate hand hygiene between patients, and between patients and workstation. Designing the base to allow direct visualization of the main treatment area.
Consulting facilities	Incorporating facilities for staff to consult episodically within the ED. Under desk brackets/shelves for computers with cable tidying systems. Including an enclosed area for clinical discussion adjacent to the main staff base.
Lighting for image viewers	Considering specific lighting requirements for the use of image viewers in the base area. All departments should be equipped with sufficient PACS viewing stations

Functional content

Spatial relationships	
• The design needs to facilitate staff flow between work station and cubicles	• Clear views of nurse call displays
Functional requirements	
• Aspects to be factored in include ergonomics, such as computer monitors being 800mm from eyes and at correct height • Bench space around computers must be adequate to allow for papers to be on the desk. Desk height is ideally adjustable • Standing Desks to improve productivity • Sufficient space to allow for an adequate number of chairs and circulation within staff area (opposing work desks should have minimum of 2m between desk edges)	• Work station(s) ideally should have clear vision of the ED clinical areas they are serving • Consider separate work stations for different areas of ED. • Consider local work desks for cubicle groups (e.g. on end of dividing wall) as this allows data entry without leaving the patient • Design needs to consider security for staff and privacy for conversations.

Specific equipment requirements

- One computer per work desk – consider the need for widescreen or dual monitors
- All monitors suitable for viewing standard radiology images
- Selected monitors suitable for viewing special radiology images (e.g. CT/MRI/USS)
- One telephone per work desk
- Senior Consultant in Emergency Medicine desk to have Direct Dial In (DDI) facility for GP calls
- Nurse in Charge desk to have DDI phone for emergency services uses
- Space for central monitoring, printers, photocopiers, nurse call system control panel
- Waste paper baskets, confidential waste collection point
- Intercom system (if fitted)
- Space for storage of patient records, stationery, reference material
- Radio aerial connectors for emergency services and internal systems (unless placed in triage areas)
- Duress alarm (probably under desk but must be located so it will not be in contact with the knees of whoever is sitting at the desk). For large department personal security devices linked to a main hub are needed.
- Pneumatic tube to access labs, consider need to access wards and departments
- Adequate power and data points (see IT and Communications section)
- Whiteboards and pin boards: consider using pin board as upright backing for work desk.

Common design flaws

- ♦ Security: Poor sightlines across the department resulting in a risk for patients & staff
- ♦ Data & confidentiality: poor placing can result in GDPR breaches.
- ♦ Spacing/configuration:
 - » Lack of adequate allowance for personal items
 - » Height of the counters/screens need careful consideration.
 - » Adequate space for PC terminals and consumables

Interview room

An interview room is used to conduct patient (including psychiatric patient) and relative and carer interviews safely, comfortably, and privately. It also accommodates the work of social workers for sensitive cases.

Size

- ♦ The minimum acceptable floor area for an interview room is 12m²
- ♦ Should comfortably fit at least three people, without feeling crowded

Functional content

Spatial relationships	
The interview room should be in a high visibility part of the ED.	Near to a treatment/examination room for physical assessment.
Functional requirements	
- Sound proof (from external auditory stimuli) - Be directly visualised by observers (one-way mirror, CCTV) - Not contain objects that may be thrown at staff e.g. light weight chairs - Furniture should have soft corners and no hard edges, and no parts of the furniture should be able to be removed and potentially used as a weapon - A functional duress/panic alarm is needed within easy reach of the interviewer and at each exit - Have low stimuli décor and muted colours - The best practice guidance for mental health from the Health Service Executive (HSE), and the Royal College of Emergency Medicine and the Colleges Psychiatry are essential resources for ensuring quality care in mental health emergencies and all designs should align	- Two doors, opening outwards and able to be locked from the outside but not from the inside - The room should be square and not have any narrow corners in which people could become cornered - The room does not require medical gases or suction - There should be no vents or lights or other apparatus that could be used as a hanging point - Smoke detector(s)
Specific equipment requirements	
A couch and/or appropriate furniture	Duress alarms.

Common design flaws

- ♦ Items in the rooms that could be used to harm participants
- ♦ A room with high levels of auditory/visual stimulation e.g. television
- ♦ A room with poor links to Triage Area

Teaching facilities

The ED requires dedicated facilities for formal education, tutorials/ /high fidelity immersive simulation teaching and meetings. This area may be used by medical, nursing, or other staff and undergraduates. It should be a private, non-clinical area with noise attenuation, often near the staff room and offices, and with access to toilets and amenities.

Size

The minimum acceptable floor area for teaching facilities is 32m² and should solely be used for teaching purposes.

Design considerations

Natural light and expandable space	Prioritizing natural light in teaching rooms. Ensuring an ability to expand the area for large groups. Large EDs may benefit from having more than one room or a dividable room with a high-quality sound-attenuating divider.
Adequate darkening and viewing equipment	Designing teaching rooms that can be adequately darkened for viewing projected material. Recognizing that projectors offer better viewing compared to large television displays.
Size considerations for maximum capacity	Calculating the size of the area based on the maximum number of people likely to use it at any one time. Considering scenarios such as combined medical and nursing meetings, immersive simulation sessions, or network teaching sessions.
Accessibility for teaching staff	Necessity for the area to be within the ED due to the potential need for senior doctors and nurses, to attend the Resuscitation area without delay while teaching. A seminar/teaching room should be situated within the administrative area of the ED.
Comprehensive teaching facilities	Teaching facilities within the ED should be comparable to those available elsewhere on site. Inclusion of audio-visual aids, computer access with large TV screens, teleconferencing hardware, PACS viewing facilities, emergency library access and storage for teaching materials.
Skills and training programs	Facilities for skills teaching and assessment are essential for delivering current educational programs. EDs providing Higher Specialist Training will require a study room and office space to accommodate Specialist Registrar trainees. Meeting the minimum requirements set down by the training body for these study and office spaces. With national training bodies moving to tele-education, consideration should be given to videoconferencing facilities within EDs and IT infrastructure that supports the delivery of hybrid teaching.

Functional content

Spatial relationships	
- The area needs to be located within the ED to ensure that senior doctors and nurses can attend the Resuscitation area promptly when teaching sessions are ongoing, but it should be separate from the clinical area. - A Seminar room should be situated within the Staff Room area of the ED.	
Functional requirements	
- Multiple sockets and access to Wi-Fi - Library for written, audio-visual and electronic reference materials. Ideally, all computers can access electronic resources	- Videoconferencing capability - Storage for simulation mannequins, teaching aids and training materials - Bookshelves for written reference materials and journals
Specific equipment requirements	
- Tables and stackable chairs - Audio-visual equipment with ability to plug in mobile devices/laptop - Ceiling/wall-mounted projection screen - Networked computers - Wi-Fi access	- Teaching aids and equipment - X-ray viewing facilities/digital imaging system - Whiteboard (electronic whiteboard preferable) - Telephone

Common design flaws

Not In good proximity, should not have to leave the ED to access teaching facilities.

Staff room

Well-designed staff facilities provide time out, relaxation and add to morale and staff functioning. The staff room is used by staff to consume meals, for social events and for the celebration of achievements. The kitchen area may be incorporated within the staff room or immediately adjacent to it. Staff should be able to prepare hot and cold drinks and prepare or heat food.

Size

- ◆ The minimum acceptable floor area for the Staff Rest and Mini Kitchen Allocation is 1.8m² per staff member on staff.
- ◆ This minimum acceptable floor area for the Staff Rest should cater to at least 1/3 of the staff on a shift with contingency.
- ◆ The size of the staff room should be big enough to seat all staff on a rostered meal break.
- ◆ Social distancing should be considered when determining these measurements.

Design considerations

Rest and relaxation facilities	At least one room should be provided within the department for staff to relax during rest periods. TV, radio, and other facilities should be available in the rest area. Provision for the preparation of food and drink should be included. Appropriate table and seating arrangements should be provided for staff comfort.
Location and access	The rest area should be located away from patient care areas. Access to natural lighting and appropriate floor and wall coverings are essential. There must be adequate access for disabled staff Staff should be able to arrive for or leave their shifts without traversing the clinical floor, along with having access to other clinical areas.
Changing and shower facilities	Adequate staff changing, locker, shower, and toilet facilities should be available and close by.
Other considerations	ABHR or hand basin at the entrance to the staff room Multiple microwaves to allow faster preparation of food Industrial grade, fast cycle dishwashers, and industrial refrigerators A managed beverage service negates the need for replenishment and maintenance of tea and coffee. A lactation room for breastfeeding staff / patients could be considered. Non-gendered spaces with separate individual changing cubicles, toilets, and showers.

Functional content

Spatial relationships	
● Toilets in close proximity ● Offices usually located in the same area of the ED.	● Located close to counselling room for staff and teaching facilities

Functional requirements	
• Preparation and consumption of meals • Located away from patient care areas • Access to an outside area if possible	• Natural lighting is desirable with secure, swipe access • Other aspects of relaxation e.g. music, television.
Specific equipment requirements	
• Kitchen/pantry • A food preparation area • Television, telephone with tables and comfortable seating (for dining and for relaxation)	• On demand hot and cold water • Microwave oven(s) • An industrial grade refrigerator and dishwasher.

Common design flaws

- ◆ A staff room that is too small with sub-optimal changing facilities/showers/lockers
- ◆ Using staff room for meetings
- ◆ Located in the middle of an ED corridor filled with trolleys.
- ◆ Notice boards, mail boxes, ED performance statistics and lockers in the staff room
- ◆ Concertina walls between staff room and adjacent meeting rooms, as these are rarely sound proof
- ◆ No interview/counselling space for Staff to debrief

Staff changing facilities

Size

- ◆ The minimum acceptable floor area for the Staff Changing Allowance is 1.4m² per staff member on staff.
- ◆ There should be a 1:2 male: female split
- ◆ A locker space of 0.5m² per staff member should be provided

Functional content

Spatial relationships
● Should be located close to clean utility areas to ensure efficient restocking
Functional requirements
● The room should be secure with access limited to authorised personnel only. There should be sufficient space and power sockets to store and charge battery powered equipment e.g. infusion pumps.
Specific equipment requirements
● As a general principle, EDs should have sufficient storage space to carry a week's worth of disposable medical supplies and intravenous fluids.

Common design flaws

- ◆ Staff toilets without restricted access (either a numeric keypad or proxy card)
- ◆ An inadequate locker facility for cyclists
- ◆ An inadequate number of shower facilities for the number of cyclists who work in the ED.
- ◆ No interview/counselling space for Staff to debrief

Store room

The equipment/store room is used for storing equipment and disposable supplies for the ED. Portable equipment may also be stored and recharged in this room.

Size

Minimum acceptable floor areas:

- ♦ linen trolley - 0.5m²
- ♦ linen store - 4m²
- ♦ mobile/hoist equipment store - 2m²
- ♦ ambulance equipment store - 8m²
- ♦ pharmacy store - 8m²
- ♦ storage solution for trolleys and wheelchairs

Design considerations

Diverse storage needs	Sterile and non-sterile supplies, stationary supplies, hazardous materials, medical gas storage, linen storage, wheelchair and trolley storage, electronic equipment storage with re-charging capabilities, clinical and non-clinical waste disposal, and Major Incident Equipment storage. Storage facilities in EDs tend to be very limited. Use of inappropriate areas for storage and unsecured spaces for materials is common. EDs must have a major incident store to carry the equipment needed in the event of a major incident. The major incident equipment storage should align with the role of the individual ED in a major incident or disaster.
Central and decentralised storage	The total area of dedicated store rooms must take into account central and decentralised storage of equipment and disposables, in conjunction with the presence of equipment at the bedside. A decentralised storage solution in large EDs including separate storage for disposables, trauma resuscitation equipment, mobile equipment, stationary A separate room for equipment servicing should be considered Bar coded systems allowing top up of supplies on a daily basis can reduce the need for larger amounts of stock to be kept in the equipment room Adequate storage facilities need to be included in any ED development or redevelopment.

Functional content

Spatial relationships

- Should be located close to clean utility areas to ensure efficient restocking

Functional requirements

- The room should be secure with access limited to authorised personnel only. There should be sufficient space and power sockets to store and charge battery powered equipment e.g. infusion pumps.

Specific equipment requirements

- As a general principle, EDs should have sufficient storage space to carry a week's worth of disposable emergency medical supplies

Common design flaws

- ◆ Poor supply chain logistics creating poor efficiency of space utilisation
- ◆ Equipment and supplies distant to where it is required to be used clinically

Administration

Offices provide space for the administrative, managerial, safety and quality, teaching, and research roles of the ED.

Size

Minimum acceptable floor areas:

- ◆ Senior Nurse's individual office accommodation - 8m²
- ◆ Multi-person office for Clinical Nurse Manager 2 grades - 12m²
- ◆ Questions around how much administration can be centralised should be addressed.
- ◆ Hot desking in an open-plan environment with meeting rooms is suggested for Consultants in EM with dedicated space for charts, books, written correspondence to reduce redundant office space.
- ◆ 4-person office - 24m².
- ◆ Open-plan office - 5m² per person.

Design considerations

General administrative area	Extensive administrative functions are crucial for supporting ED clinical work. Archiving of clinical material, clinical audit, preparation of medical reports, patient discharge support, and liaison with primary care and other services.
Dedicated office space	A fully equipped office for designated administrative and secretarial staff is required. This area should be adjacent to the ED. Secretarial accommodation should reflect the need for each Consultant in Emergency Medicine to have a dedicated secretary.
Secretarial accommodation requirements	Shared secretarial support may become feasible with significant increases in Consultant numbers and changes in work practices. Consultants in Emergency Medicine tend to currently have individual offices. There is the potential for potential future use of shared office spaces and hot desking with extended working hours and out-of-hours rostering.
Consultant office specifications	Individual offices are essential for confidential matters such as staff appraisal, mentoring, counselling, risk management, and handling complaints, along with small and larger consultation rooms for interviewing staff and management/governance meetings.
Other considerations	The total office space should be sufficient for future expansion of staff numbers and new roles There is an emerging trend for common and shared office areas Some hospitals have common meeting areas for all departments to share

Telemedicine	EDs should have telemedicine facilities with a dedicated room with appropriate communications infrastructure and equipment. This room should be of a suitable size to allow for simultaneous viewing by members of multiple service teams.
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Functional content

Spatial relationships	
Should be close to reception area	
Functional requirements	
- Offices should be located close to each other and close to secretarial services but not within the clinical area - There should be ready access to printing and scanning/photocopying - The area should be accessible by authorised staff only	- Wi-Fi should be available throughout - Staff members who are mostly office workers should, if possible, have offices with natural light - The administrative area should have room(s) in which private meetings may be held.
Specific equipment requirements	
n/a	

Common design flaws

- ◆ Lack of reporting areas
- ◆ Should have staff access to reporting area located adjacent to each area for input into Electronic Patient Records (EPR).

Distressed relatives' room / quiet room

A distressed relatives' room provides a private, quiet room for distressed relatives, carers or friends of all ages and cultural backgrounds to gather. The room also provides a space to obtain a collateral patient history and potentially to deliver bad news about a patient.

Size

- ♦ The minimum acceptable floor area for a Quiet Room is 16m²
- ♦ It should be able to comfortably seat at least four relatives, friends or carers as well as seating room for a nurse or doctor to enter and deliver news regarding a patient.

Design considerations

Safety measures	Panic button: a panic button should be installed for emergency situations. Exit routes: two clear exit routes must be available for quick evacuation. Fixed furniture: furniture and fittings should be fixed to prevent their use as weapons.
Aesthetic considerations	Calm colours: decorate the room in calm colours to create a soothing environment. Well-lit: ensure good lighting using non-breakable glass fixtures to maintain safety. Calming, peaceful music may or may not be appropriate Cultural or religion specific needs merit consideration
Visibility and observation	Observation window: include an observation window for monitoring. External view: design the room to be viewable from outside for enhanced surveillance.
Patient interview and access control	In small EDs, this room could double as a psychiatric interview room, however ideally, they should be separate due to the unique requirements of each Unrestricted access to waiting room, but controlled access to the ED.

Functional content

Spatial relationships	
• The room should be close to the resuscitation area. • Refreshment facilities should be integrated into the design, such as a small kitchenette-style setup. • Ready access to toilets	• The room should not be so close that these people can see or overhear clinical care being delivered to the patient (or to other patients).
Functional requirements	
• Private and soundproof space • Where distressed relatives, friends or carers may gather and support each other	• Culturally neutral and inoffensive • Aesthetically calming and peaceful.
Specific equipment requirements	
• Comfortable furnishings	• Power point(s) for mobile phone charging

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|--|--|
| <ul style="list-style-type: none"> • A telephone with mobile and external line access | <ul style="list-style-type: none"> • Muted décor and unobtrusive, culturally non-offensive artwork. |
|--|--|

Common design flaws

- ◆ Cramped rooms with inadequate seating or natural light
- ◆ Inappropriately locating the distressed relatives' room near busy clinical areas.
- ◆ Rooms that allow for interruptions in the form of external noise or other people traversing the area
- ◆ A space that has been purpose built for something else e.g. treatment cubicle, an isolation room
- ◆ Rooms with restricted access to communication e.g. no telephone line or a mobile phone black spot
- ◆ A space that is aesthetically too stimulating or confronting

Viewing room

A viewing room in the ED is a private, adaptable space located near the resuscitation area, providing grieving relatives with privacy to view deceased patients and observe diverse religious practices.

Size

Minimum acceptable floor areas:

- ◆ Viewing Room - 13.5m²
- ◆ Viewing Room anteroom - 5m².

Design considerations

General	
	Entrance to Viewing Room does not require one to travel through the ED floor

Functional content

Spatial relationships

- Proximity to Resuscitation Room: the private viewing room is strategically located near the resuscitation room for a discreet transfer of deceased patients
- Adjacency to Relatives' Room: in proximity to the relatives' room ensures privacy for family and friends entering and leaving the viewing area
- Privacy Considerations: Spatial design ensures the deceased is transferred out of sight of other patients, maintaining sensitivity and respect.

Functional requirements

- | | |
|---|--|
| <ul style="list-style-type: none"> • Designed to accommodate diverse religious denominations, allowing for rituals and customs to be observed • Functional layout to prioritise privacy, providing a space where relatives can grieve, and view remains without intrusion | <ul style="list-style-type: none"> • Functional efficiency supports a smooth and private transfer of the deceased, minimising disruption to both medical staff and grieving families. |
|---|--|

Specific equipment requirements

- With furnishings and lighting appropriate to create a comforting environment for grieving relatives during the viewing
- Furnished to facilitate the respectful and dignified transfer of the deceased, ensuring a seamless process
- Provision of necessary equipment to cater to diverse religious practices, such as storage for religious items or fixtures for specific ceremonies.

Common design flaws

- ♦ Poor Signage and Wayfinding
- ♦ Limited Privacy

Strategic Planning

Patient centred care

Sub-wait areas

Consider implementing sub-wait areas for patients awaiting diagnostic imaging, test results, team assessments, discharge planning, or transport. These spaces can also accommodate patients needing supervision or separation, such as those in Garda custody. Directing patients to specific rooms from the Front Door can minimise central waiting areas, prompting a re-evaluation of traditional reception desks into information resources. Given that 25–30% of ED patients are children, planning teams should establish a dedicated children's flow system and area. This waiting space should ensure staff observation without allowing adult patients or visitors to view children waiting (audio-visual separation of adult patients from children). This is also true for specialist pathways such as Mental Health and Geriatric Medicine.

Multiple sub-wait areas should also be established, starting from the post-triage space and extending to ambulatory/minors areas, majors, Mental Health, and Frail Elderly areas, and located beside the diagnostics hub (CT/X-Ray). This is crucial in ED design to reduce congestion, improve patient flow, and cater for the diverse needs of patients presenting to the department.

Patient toilet and shower facilities

In the design of an ED, it is essential to ensure that patients in waiting and clinical areas have convenient access to toilets. Baby changing facilities must be provided for families. Each set of 12 treatment rooms/bays should include a dedicated patient shower area, and mandatory disabled toilet facilities should be incorporated.

To enhance safety, all toilets must be equipped with alarms, enabling emergency access and patient retrieval in the event of a collapse. To mitigate the risks associated with anti-social behaviour or substance misuse in toilet areas, patient-accessible toilets should be well-equipped with features such as appropriate light fittings, secure ceilings, and immovable fixtures. Moreover, en-suite toilets are required for isolation rooms, the gynaecology room and any space designated for forensic examination of victims of sexual violence.

Refreshment supply zone for patients

Within the ED, create a dedicated space to provide basic refreshments for patients and their relatives, easily accessible from the waiting area. This area, potentially designed as a coffee station or similar, must strictly comply with national health, safety, and food preparation guidelines.

Geriatric Emergency Medicine space (GEMS)

The increasing number of older individuals attending EDs reflects the demographic shift marked by a rapid rise in the aging population. Projections indicate that from 2015 to 2030, the share of the population aged 65 and over is expected to increase from one in eight to one in six, with the number of people aged 85 and over almost doubling. Older patients presenting to the ED pose unique challenges, showing a higher likelihood of admission and a longer length of stay compared to other patient groups. Addressing the needs of increasingly frail older individuals necessitates a comprehensive, whole-system approach to emergency care, with a pressing need for improved quality, outcomes, and efficiency. As such, the entire adult ED needs to be age attuned to cater for the older adult demographic.

Geriatric Emergency Medicine (GEM) Units represent a specialised response to these challenges, aiming to enhance the experience and outcomes of older adults in EDs. The GEM Unit streams eligible patients at triage and, following a thorough assessment, explores alternative care options such as home care or community settings. The GEM space is required to be age attuned to reduce falls and delirium while in ED. Hand rails, non-slip flooring, beds rather than trolleys, appropriate dimmable lighting, natural light, noise reduction ceilings and floors, arm chairs rather than chairs, clocks on walls etc. all help mitigate these issues.

Paediatric Emergency Medicine (PEM)

Specific requirements for PEM are the need to audio visually separate children from adult patients. As such this requires separate medication/clean room/infusion preparation room, family room, dirty utility etc.

Facilities must be family friendly with breast feeding/bottle warming/nappy changing facilities and space for buggies etc.

Play and distraction areas and devices help reduce anxiety and stress.

Connectivity with the adult department is a key safety feature assuming co-dependencies exist.

There are safety advantages to co-locating a Paediatric Resuscitation area in the adult resuscitation area to ensure staff support.

The space calculator can be used to determine the number of majors cubicles and ambulatory spaces needed based on activity, length of stay, hours of occupancy etc.

A 'sensory room' is important in support of neuro-diverse children.

A sedation/Procedure room as per earlier section is needed.

Bariatric medicine

The rising prevalence of obesity poses significant challenges for healthcare systems in worldwide. ED design must address the unique physical and emotional needs of obese patients while ensuring a safe and efficient work environment for staff. This encompasses planning for patients weighing up to 400kg.

Entry

- ◆ Ambulance Bay Requirements: Refer to [Appendix 2](#) for dimensions.
- ◆ Doorways: Accommodate bariatric wheelchairs and trolleys.
- ◆ Handrails: Provide support for bariatric patients.
- ◆ Corridors: Ensure adequate width for bariatric patient trolleys, associated equipment, and foot traffic passage.

Waiting room

Allocate 20% of waiting room furniture capable of supporting larger patients, integrating it with standard-sized seating to avoid stigmatization.

Patient treatment spaces

Each major ED clinical area should be equipped with the necessary capacity, space, and design to accommodate patients with bariatric medicine needs. These include:

- ◆ Resuscitation bay
- ◆ General cubicle
- ◆ Isolation/negative pressure room.

Design these rooms to accommodate a bariatric patient trolley/bed with sufficient circulating volume.

Bariatric toilets

- ◆ Floor-mounted toilets
- ◆ Adequate room on each side for assistants
- ◆ Reinforced walls to support grab rails
- ◆ Sink capable of handling additional load

Equipment

- ◆ Ceiling-Mounted Lifting Devices: Consider at least one dedicated bariatric treatment area. Alternatively, provide portable lifting machines with suitable storage space.
- ◆ Weigh Scale and Hoisting Device
- ◆ Adjunctive Specialist Equipment: e.g., slide mats

Bariatric patient movement

- ◆ Plan reception and patient pathways within the ED.
- ◆ Designated routes with clear corridors, viewing mirrors, and floor coverings suitable for bariatric patient movement.
- ◆ Consider storage and availability of equipment for patient transfers.

Healthy working environment for staff

In contrast to patients with relatively short stays, staff members dedicate extended periods, spanning months and years, within the EDs. The design of the ED environment plays a crucial role in fostering a healthy workplace for staff and necessitated thoughtful considerations. Key factors contributing to staff wellbeing include:

- ♦ **Seating facilities:** ensuring staff have designated seating areas at their bases and patient bedside, crucial for all staff.
- ♦ **Ergonomically friendly design:** prioritizing designs that promote comfort and minimise physical strain.
- ♦ **Ready access to storage:** providing convenient access to storage for equipment and supplies to facilitate seamless workflow.
- ♦ **Flexible work options:** offering staff the flexibility to choose where and how they work, including the option of standing or sitting at staff bases.
- ♦ **Crowd limitation and circulation space:** designing the workplace to prevent overcrowding, allowing for adequate circulation space.
- ♦ **Quiet discussion areas:** incorporating designated areas for quieter staff discussions throughout the ED.
- ♦ **Noise control and attenuation:** implementing measures to control and attenuate noise levels within the environment.
- ♦ **Air quality:** addressing air quality concerns, including thermal comfort, ventilation, and the dispersal of offensive smells.
- ♦ **Access to outside views and daylight:** incorporating elements that connect staff with the outdoors, enhancing morale.
- ♦ **Low-level lighting at night:** providing suitable lighting to assist staff in maintaining diurnal body rhythms during night-time shifts.
- ♦ **Separate staff facilities:** designing dedicated staff facilities physically separated from clinical areas to enable proper breaks from patient care and exit access separate from that used by patients.
- ♦ **Spacious staff rooms:** ensuring staff rooms can accommodate the maximum number of users during meal breaks, equipped with chairs, food preparation areas and microwaves.
- ♦ **Facilities for physical activity:** providing amenities like bike storage, showers, and lockers to encourage walking, running, and cycling.
- ♦ **Quiet Room:** Ideally be technology-free zones to provide staff with a break from screens, blue light, and device use, allowing them to sit, stand, or lie down as needed.

Infectious disease management

The Covid-19 pandemic has significantly impacted the delivery of healthcare services, prompting the need for transformative measures in urgent and emergency care¹². To address the challenges posed from infectious diseases, especially in EDs, certain design considerations are essential to ensure the safety of both patients and staff.

Maintaining social distancing and infection control measures

- ◆ ED infrastructure must be upgraded to allow social distancing for both staff and patients¹³.
- ◆ Strategically placed sub-wait areas throughout the department should be utilized to prevent congestion and overcrowding in the main waiting room, aiding in social distancing.
- ◆ In all major clinical and non-clinical areas, it is important to adhere to national policies regarding what constitutes adequate social distancing.
- ◆ Design should facilitate segregation and isolation areas for individuals with infectious diseases.
- ◆ Adequate staffing levels are crucial to compensate for the additional time required in certain patient encounters involving personal protective equipment (PPE).
- ◆ Rapid access to necessary supports, such as diagnostics, is vital for effective decision-making, minimising the time patients spend in the ED.

Preventing ED crowding

- ◆ Crowding leads to higher in-hospital mortality rates for patients and excess deaths at 30 days (Jones S, EMJ 2022).
- ◆ Crowded conditions increase the risk of adverse events for patients.
- ◆ Reducing ED crowding can help lower the median length of stay in the hospital.
- ◆ Better flow and less waiting time contribute to improved overall patient health outcomes and satisfaction.
- ◆ Effective crowd management ensures optimal use of medical staff, equipment, and facilities.
- ◆ Preventing crowding allows for timely diagnosis and treatment, critical in emergency care.
- ◆ Less crowded environments decrease stress and burnout among healthcare providers.
- ◆ Efficient patient flow enhances the overall operational efficiency of the ED.
- ◆ Maintaining manageable patient volumes helps in adhering to healthcare safety and quality standards.
- ◆ Unoccupied beds must always be available to enable immediate admission decisions for patients requiring hospital beds.
- ◆ Adequate isolation areas on wards should be in place to prevent patients from being retained in the ED solely due to infection control concerns¹⁴.
- ◆ The Patient Experience Time (PET) KPI, aiming for 95% of patients spending less than 6 hours in the ED, must be transparently implemented to avoid 'trolley waits.'
- ◆ According to Roussel et al, patients who spent the night in the ED had a higher in-hospital mortality rate (15.7% vs. 11.1%), a greater risk of adverse events compared to those admitted to the ward, and a longer median length of stay (9 days vs. 8 days)¹⁵

Enhancing community care pathways

- ◆ Consolidate and enhance pathways for treating patients outside the hospital setting.
- ◆ Large numbers of patients should not be expected to gather in ED waiting rooms, especially with the risk of nosocomial infection.
- ◆ EDs should focus on being the best place to deliver appropriate care at the right time for the right patient, rather than the only available option.
- ◆ System reform should be universal, ensuring that EDs are not solely relied upon to address failures in community and specialty care.

These guidelines emphasize the need for EDs to adapt to the post-pandemic landscape, prioritizing patient and staff safety while ensuring efficient and effective care delivery. The principles of infection control, prevention of ED crowding, and enhancement of community care pathways are pivotal in guiding the design and operation of EDs in the ongoing and future healthcare scenarios.

End-of-Life Suite (EOL)

Purpose and Distinction:

- ◆ The EOL care suite serves patients who are actively dying but have not yet passed away. It provides acute palliative care.
- ◆ Distinct from other clinical rooms, the EOL suite is a place of compassion, dignity, and support.

Location and Proximity:

- ◆ Place the EOL suite near a Viewing Room. This proximity allows families to transition seamlessly between the two spaces.
- ◆ Consider locating it near Resus (resuscitation) or Majors, as these areas are central in the ED. However, ensure that the suite has its own discreet access, separate from the main ED floor.
- ◆ The suite should have its own entrance, avoiding the need for visitors to traverse the busy ED floor.

Design Elements:

- ◆ Soundproofing and visual barriers are essential. Create a serene environment with calming colours and materials.
- ◆ Provide comfortable chairs or sofas for family members. A reclining chair for patients can enhance comfort.
- ◆ Set up a small beverage bay nearby, allowing families to have tea or coffee without leaving the suite.
- ◆ Use soft lighting and avoid clinical aesthetics. Incorporate calming artwork or nature scenes.
- ◆ Equip the suite for procedural sedation if needed. Ensure necessary equipment is readily available.

The EOL care suite is more than just a physical space; it's a place where compassion, dignity, and support intersect. By following these guidelines, hospitals can create an environment that eases the emotional burden for patients and their loved ones ¹⁶.

Storage Solutions

Efficient storage not only optimizes space but also ensures quick access to critical equipment during emergencies. Here are some strategies to consider:

Central Staff Hub in each zone as Storage Base:

- ♦ **Concept:** Promote the use of clever design concepts that save space while ensuring accessibility. Consider utilizing the base of the centrally located staff hub for emergency equipment storage.
- ♦ **Implementation:** Install secure cabinets, procedure trolleys within the staff hub. This central location allows rapid retrieval during critical situations.
- ♦ **Benefits:** Easy access, efficient organization, and proximity to staff.

Non-Linear Design:

- ♦ **Concept:** When designing storage arrangements for EDs, consider moving away from linear layouts. Instead, embrace a circular design that allows everything to be viewed more easily. By avoiding long lines with a hub at the end, this approach enhances visibility, accessibility, and overall efficiency within the ED space.
- ♦ **Implementation:** Use modular shelving, rotating storage units, or vertical storage solutions. These allow flexible configurations and adaptability to changing needs.
- ♦ **Benefits:** Efficient use of space, reduced clutter, and improved workflow.

Fixed vs. Mobile Storage:

- ♦ **Concept:** Evaluate the balance between fixed and mobile storage solutions.
- ♦ **Implementation:**
 - » **Fixed Storage:** Install permanent cabinets, shelves, and wall-mounted units for items that remain in specific locations.
 - » **Mobile Storage:** Use wheeled carts, portable bins, or modular units for items that need to move around the ED.
- ♦ **Benefits:** Fixed storage ensures consistency, while mobile options enhance flexibility.

Pick & Go System:

- ♦ **Concept:** Implement a system where staff can quickly “pick” necessary supplies and “go” to the patient’s location.
- ♦ **Implementation:** Organize supplies in clearly labelled bins or drawers. Staff can grab what they need without wasting time searching.
- ♦ **Benefits:** Efficient, time-saving, and reduces disruptions during patient care.

ICT and planning

Information and communications technology

The integration of Information and Communications Technology (ICT) or Information Technology (IT) stands as a pivotal force driving transformative advancements in ED services. The multifaceted application of ICT in the ED setting can be categorised into five key areas: Clinical infrastructure, Communications, Administrative functions, Facilities management and security, and Patient engagement.

Efficient clinical operations hinge on a robust infrastructure, where a unified communications cabling system for voice, data, nurse call, security, and closed-circuit television is vital.

Adequate accessibility to electronic resources is ensured through desktop computers strategically placed between cubicles and an allocation of one computer with a desk phone per 1200mm of staff base width. With the introduction of hand-held tablets, this may be changed.

Additional considerations encompass power outlets, network/wifi signal booster points, and designated areas for shift leaders equipped with essential communication tools. Moreover, printers are strategically placed, promoting standardised paper types and forms, with printing directed to a server for secure and reliable functionality.

Intercom/announcement technology should be avoided.

Nurse call and emergency assistance systems must be prioritised.

Ensuring strong mobile phone signal strength, and effective telephone communication in patient treatment spaces become imperative.

Mobile communication devices for staff, including DECT phones and voice-activated devices, should ideally be unified through a communications integration platform. The inclusion of fixed and mobile duress alarms, Warden Intercommunication Point (WIP) phones, and walkie-talkies addresses varied communication needs, particularly in mass casualty or disaster situations.

Administratively, the availability of computers for support staff is critical, with a preference for a computer per workstation, whether desktop or laptop.

Ensuring secure access to electronic information for clinical staff working remotely is facilitated through Virtual Private Networks (VPN) or similar methodologies.

Fixed and portable duress alarms, robust closed-circuit television systems in key areas such as entrances/exits, waiting rooms, and ambulance bays contributes to staff safety and the monitoring of patient activity.

The patient experience is enhanced by accessible phone charging stations, positive distractions like music and television, internet access, real-time public transport scheduling screens, such as for local buses, etc.

Anticipating the transition to electronic medical records, minimal joinery/storage for paper records is recommended. Centralised communication hubs, VoIP integration, and investment in an integrated communications platform are suggested for future-proofing EDs.

Common pitfalls, including inadequate anticipation of computer space requirements and a segregated approach to IT implementation, are highlighted. The current trend emphasizes the need for networked diagnostic sets and physiological monitoring, with an eventual shift toward one computer per cubicle to accommodate fully digital electronic records.

Handheld tablets

These require a climate controlled area for charging with enough electric supply for all tablets.

Docking stations for charging handheld tablets used in patient care may be considered. Ensure the number of stations is proportional to the staff on duty, with a designated area per staff member for convenient access.

Future technologies

3D printing, remote robotics, service robots, AI, and advanced radiological practices will impact future healthcare delivery to varying levels.

Telemedicine

Integrating strategic bed management systems empowers staff to efficiently coordinate capacity.

Facilitating pre-alerts to the ED from ambulance/paramedic networks through electronic noticeboards for timely patient assessments will facilitate streaming, early intervention, redirection, non-conveyance/alternative pathway decisions.

EDs using telemedicine facilities should have a dedicated room with appropriate communications infrastructure and equipment. This room should be of a suitable size to allow for simultaneous viewing by members of multiple service teams. It is important to consider its adjacency to the central staff hub/clinical manager's area.

Major incident planning

ED design must undergo scrutiny to ensure suitability for Major Incident response. It necessitates designated areas for mass triage and the ability to reconfigure clinical spaces for critically ill or injured patients. Effective communication with hospital control areas becomes crucial and local Major Incident plans should be incorporated during the ED design phase. Consideration needs to be given to adequate and appropriate storage space for Major Incident equipment, taking into account adjacency to ambulance/walk-in entrance points and proximity to critical care areas.

Art in ED

In the ED, incorporating art can contribute positively to the environment, enhancing the overall patient experience. It is essential that the chosen artwork aligns with local infection control and health and safety policies. Seeking guidance on the optimal utilization of art in a clinical setting is advised to ensure its beneficial impact. Additionally, considering community engagement for tailored art reflecting the local population and culture can further enrich the healing atmosphere within the ED.

Patient pathways and required adjacencies (adult, paediatric and staff zones)

The following diagrams are not design options or developed layouts, but demonstrate the arrangements and adjacencies that could be suggested within the ED.

Figure 1 - Adult zone required adjacencies

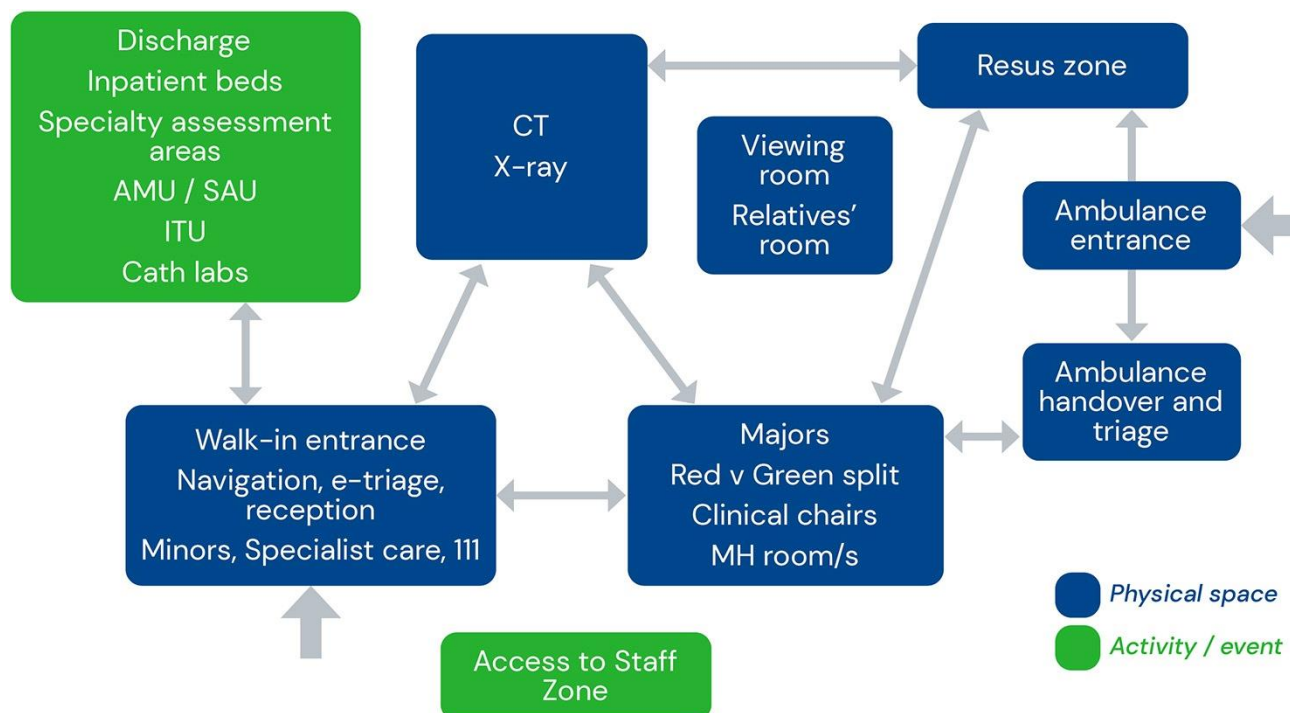


Figure 2 - Paediatric zone required adjacencies

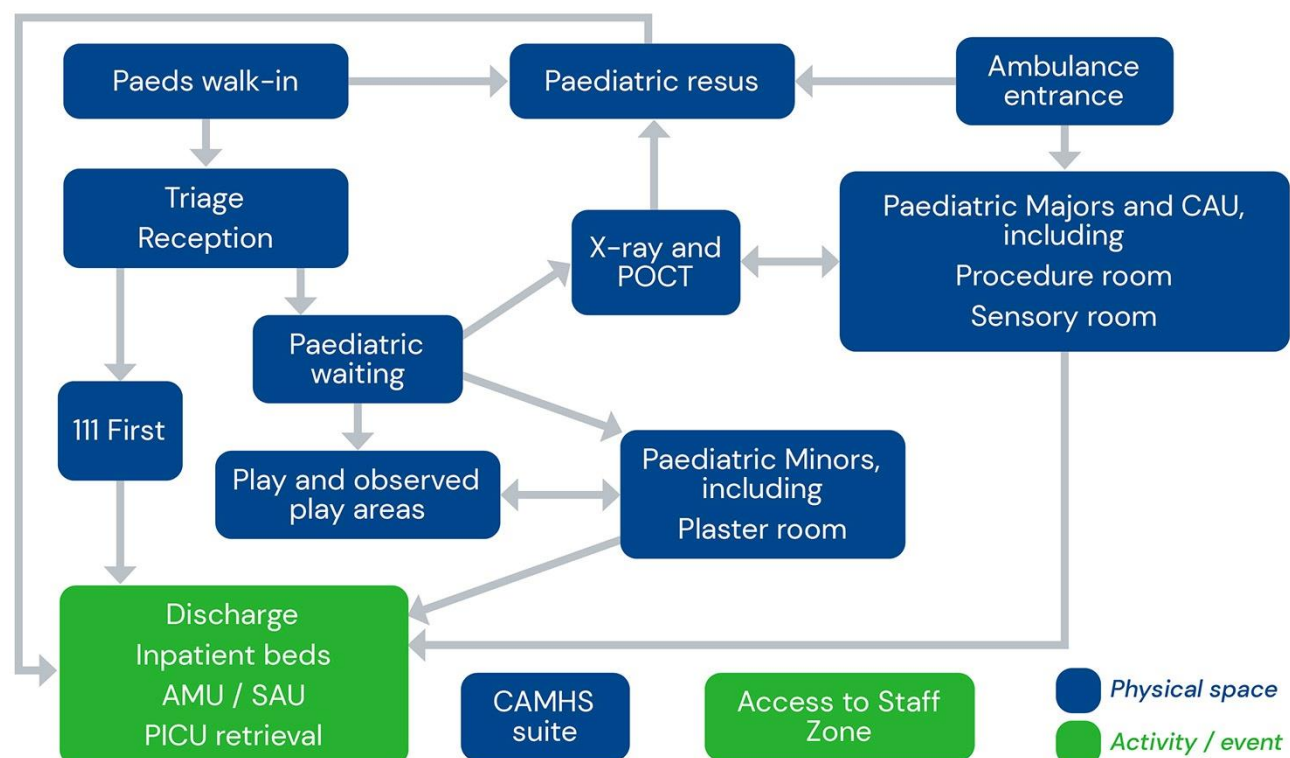


Figure 3 - Staff zone required adjacencies



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Appendices

[Appendix 1: Summary of description, size and common flaws](#)

[Appendix 2: Summary of Functional Content by room type](#)

[Appendix 3: Summary of Design Considerations by room type](#)

[Appendix 4: SOA](#)